Section I

VASCULAR ACCESS

He who works with his hands is a laborer.
He who works with his head and his hands is a craftsman.

Louis Nizer
Between You and Me
1948

Chapter 1

Vascular Access Primer

It is not a bad definition of man to describe him as a tool-making animal.

Charles Babbage (*a*)

INTRODUCTION

One of the most dramatic events in medical self-experimentation took place in a small German hospital during the summer of 1929, when an intrepid surgical resident named Werner Forssmann inserted a plastic urethral catheter into the basilic vein in his right arm and advanced the catheter into the right atrium of his heart (1). This was the first documented instance of a right-heart catheterization in a human subject, but Dr. Forssmann received no accolades, as he had acted in defiance of the senior surgical staff at his hospital. Instead, he was promptly dismissed from his residency for actions that were perceived as inappropriate and reckless. Upon dismissal, he was told that "such methods are good for a circus, but not for a respected hospital" (1). Dr. Forssmann went on to become a country doctor, but his achievement in vascular cannulation was finally recognized in 1956, when he was awarded the Nobel Prize in Medicine.

Werner Forssmann's achievement was possible because he used a flexible plastic catheter that could safely follow the contours of the venous system as it was advanced. This was a departure from the traditional practice of cannulating blood vessels with rigid needles and metal cannulas, and it heralded the modern era of vascular cannulation, which employs a wide array of flexible plastic catheters like the ones described in this chapter.

CATHETER BASICS

Vascular catheters are made of synthetic plastic polymers that are chemically inert, biocompatible, and resistant to chemical and thermal degradation. Catheters that are used for short-term cannulation (days to weeks) are typically made of *polyurethane*, a versatile polymer that is pliable, yet provides enough tensile strength to resist kinking during catheter insertion. (The elastic fibers used in stretchable clothing are made of polyurethane.) Catheters that are used

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for long-term vascular access (months) are typically made of *silicone*, which is much more pliable than polyurethane (e.g., the nipple on baby bottles is made of silicone) and is less likely to cause vascular damage. Because of their pliability, silicone catheters are difficult to insert percutaneously, and are used primarily as implantable catheters (such as those used for long-term chemotherapy).

Catheter Size

The size of vascular catheters is a reflection of the *outside diameter* of the catheter. There are two expressions of catheter size: gauge size and French size. The correlation between the two is shown in Table 1.1.

Gauge Size

The gauge system was introduced (in England) for sizing solid iron wires, and was later adopted for hollow needles and catheters. Gauge size is a measure of how many wires (or catheters) can be placed side-by-side in a given space, and it varies inversely with outside diameter (OD); i.e., the higher the gauge size, the more catheters will fit in a given space, and thus the smaller the OD. Unfortunately, the actual OD for each gauge size is not standardized, and varies with each manufacturer. Gauge sizes are typically used for needles, small-bore single-lumen catheters, and the infusion channels in multilumen catheters; sizes typically range from 16 gauge (largest diameter) to 21 gauge (smallest diameter).

French Size

The French system was introduced (guess where) for sizing hollow tubes (catheters), and it provides a more predictable measure than the gauge system. The French scale begins at zero, and each increment of one French unit represents an increase in OD of 0.33 millimeters (2): i.e.,

$$\text{French size (Fr)} \times 0.33 = \text{OD (mm)}. \tag{1.1}$$

French sizes are used for multilumen catheters and large-bore single-lumen catheters; sizes typically range from 4 Fr (1.2 mm OD) to 9 Fr (3.2 mm OD). French sizing is also used for urinary catheters, nasogastric tubes, and pleural drainage tubes.

Flow Through Catheters

The determinants of flow through narrow, rigid tubes (e.g., catheters) was first described by a German civil engineer (Gotthilf Hagen) and a French physician (Jean Marie Poiseuille), working independently in the mid-19th century. Their observations are expressed in the following equation, known as the *Hagen-Poiseuille equation* (3).

$$Q = \Delta P \times (\pi r^4/8\mu L). \tag{1.2}$$

This equation states that the steady or laminar flow rate (Q) in a rigid tube is directly related to the pressure gradient along the tube ($\Delta P = Pin - Pout$) and the fourth power of the radius of the tube (r^4), and is inversely related to the length of the tube (L) and the viscosity of the fluid (μ). The term enclosed in parentheses is equivalent to the reciprocal of resistance (according to the relationship: $Q = \Delta P \times 1/R$), so the resistance to flow can be expressed as: $R = 8\mu L/\pi r^4$.

TABLE 1.1	Correlation between French and Gauge Sizes	
French Size	**Gauge Size**	**Outside Diameter**
1	27	0.4 mm
3	20	0.9 mm
4	18	1.2 mm
5	16	1.7 mm
7	13	2.4 mm
9	11	3.2 mm

From Access Technologies, 2016 catalogue (available at www.norfolkaccess.com).

Catheter Dimensions

The Hagen-Poiseuille equation describes the influence of catheter dimensions on flow through the catheter. This is an important consideration because *the infusion rate of intravenous fluids is determined by the dimensions of the indwelling catheter, and not by the size of the cannulated vein*. The inner radius of the catheter has a profound influence on flow (because flow rate is directly related to the fourth power of the radius), and this is demonstrated in the bar graph on the left in Figure 1.1 (4). In this case, the gravity-driven flow of water through a 16 gauge catheter was more than double the flow through an 18 gauge catheter, and was almost four times greater than the flow through a 20 gauge catheter (with all catheters being equal in length). The large difference in flow rates between 16 and 20 gauge catheters is associated with less than a one millimeter difference in outside diameter (see Table 1.1), which highlights the importance of catheter diameter as a determinant of flow.

The Hagen-Poiseuille equation also shows that flow will vary in an opposite direction to changes in the length of a catheter; this is demonstrated in the graph on the right in Figure 1.1 (5). Note that the transition from a two-inch catheter (a common length for peripheral vein catheters) to a six-inch catheter (a common length for central venous catheters) is associated with a 40% reduction in flow, and a further transition to a 12-inch catheter (an available length for central venous catheters) is associated with an additional 40% decline in flow.

The information just presented indicates that *when rapid volume infusion is needed, a large-bore catheter is the appropriate choice, and a short, large-bore catheter is the optimal choice.*

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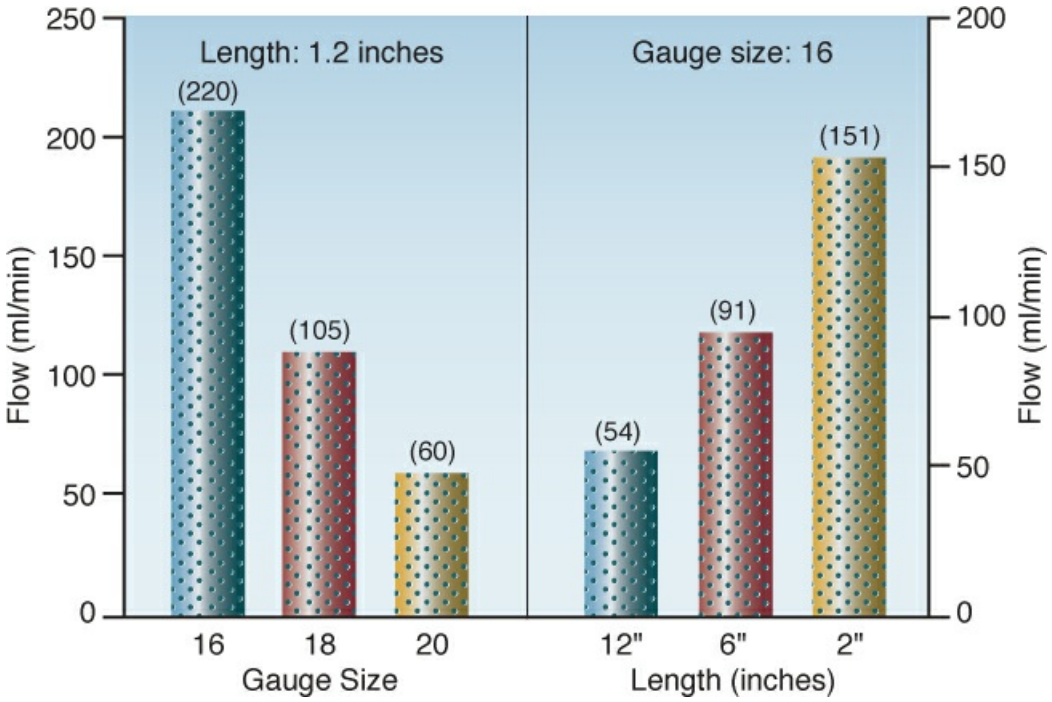


FIGURE 1.1 Graphs demonstrating the influence of catheter diameter (panel on the left) and catheter length (panel on the right) on flow rate. See text for further explanation. Data from References 4 and 5.

Infusion Pressure

The resistance to flow created by vascular catheters can be overcome by increasing the pressure gradient for flow (i.e., the ΔP in Equation 1.2). In a gravity-driven infusion system, this is accomplished by increasing the height of the infusate container (bag or bottle) above the cannulation site: e.g., a height of 68 cm (27 inches) will create an infusion pressure of 50 mm Hg (or one pound per square inch, psi), and an increase in height to 100 cm (39 inches) will increase the infusion pressure to 75 mm Hg (1.5 psi) (6).

INFUSION PUMPS: Intravenous fluids and drugs are typically delivered at specific infusion (or dose) rates, and this control is achieved with programmable infusion pumps that adjust the infusion pressure (up to 15 psi) to deliver a preselected infusion (or dosage) rate. There are two types of infusions pumps: *volumetric pumps*, which can deliver one liter of fluid (from a bag or bottle) at flow rates of 0.1 to 1,000 mL/hr, and *syringe drivers* that operate with a lower volume (up to 100 mL) and deliver fluid at rates of 0.1 to 100 mL/hr (6). *Volumetric pumps* are general-purpose devices that are used to deliver intravenous fluids and most intravenous drugs, while syringe pumps are popular for patient-controlled analgesia. There are also specialized infusion pumps for the resuscitation of massive hemorrhage. These devices deliver warmed fluids or blood products at rates of up to 1.5 L/min when combined with specialized “rapid infusion catheters” that are typically 7–8 French in diameter and 2–2.5 inches in length.

GENERAL-PURPOSE CATHETERS

Intravenous catheters are classified as central or peripheral catheters based on the following simple distinction: central catheters extend into one of the vena cavae, and peripheral catheters do not. The following is a description of the peripheral and central catheters that are used in everyday patient care in the ICU. Catheters with a specialized function, like hemodialysis catheters and pulmonary artery catheters, are described elsewhere in the book. (Chapter 8 is devoted entirely to the pulmonary artery catheter.)

Peripheral Catheters

Peripheral catheters are typically inserted into one of the veins in the upper extremity, and they do not extend beyond the shoulder. Three distinct types of peripheral catheter have been identified (7,8): short peripheral catheters (<6 cm in length), long peripheral catheters (6 to <15 cm in length), and midline catheters (15 to 20 cm in length).

Short Peripheral Catheters

The traditional method of cannulating peripheral veins involves a 16–22 gauge catheter that is 3–5 cm (1–2 in) in length, and is placed in a visible or palpable vein (usually in the upper extremity) using a catheter-over-needle device like the one in Figure 1.2. The tip of the catheter is recessed back from the tip of the introducer needle, and is tapered to prevent fraying as the catheter is advanced into the blood vessel. When the tip of the probe needle enters the vein, a "flashback" of blood appears in the clear hub of the needle. When this occurs, the catheter is advanced over the needle and into the lumen of the blood vessel.

Cannulation with short peripheral catheters is favored because it provides rapid vascular access (when a superficial vein is visible or palpable), although the initial cannulation attempt fails in about one-third of cases (9). The major disadvantage of short catheters is their limited "dwell time" (the time an indwelling catheter remains functional); the reported failure rate of these catheters is 40–60% within the first 3 days (8). Common causes of failure include localized phlebitis, catheter occlusion or dislodgement, and vascular perforation with extravasation of infused fluids. Short peripheral catheters are especially problematic in ICU patients, who are often agitated and prone to dislodging these catheters.

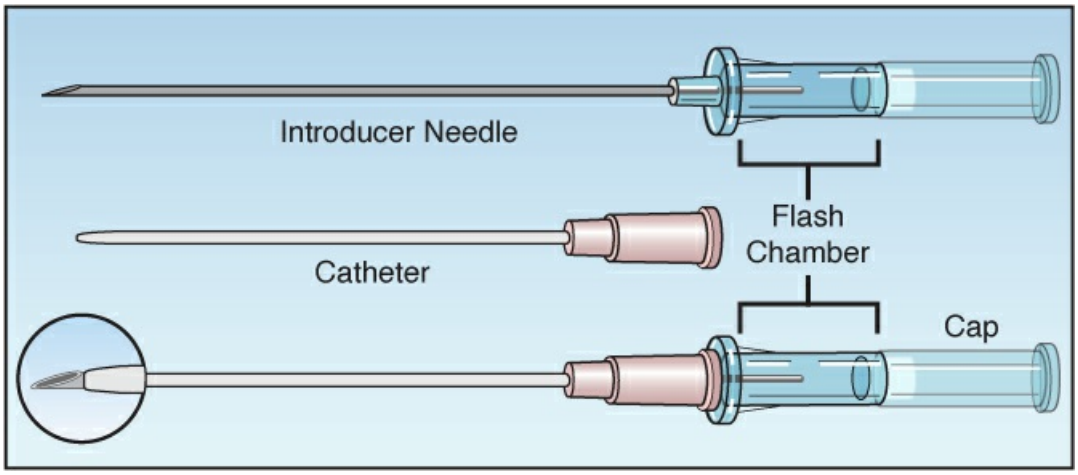


FIGURE 1.2 A catheter-over-needle device used to cannulate superficial peripheral veins.

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Long Peripheral Catheters

Long peripheral catheters (also known as “extended dwell catheters”) are typically 8 cm (3.1 inches) in length, and were introduced to improve the stability and dwell time of the short peripheral catheters. Clinical studies have consistently shown a longer dwell time with the longer catheters (about 7–9 days) (10). Despite this advantage, long peripheral catheters have not been embraced (at least not in North America), possibly due to the growing popularity of midline catheters (see next).

Midline Catheters

Midline catheters are the longest of the peripheral vein catheters (15–20 cm in length), and are inserted into one of three deep veins above the antecubital fossa: the basilic, brachial, or cephalic veins (see Figure 1.3). The basilic vein is preferred because it runs a direct course up the arm, and is not in close proximity to an artery or nerve (like the brachial vein). Because the major veins in the upper arm are deeply situated ultrasound guidance is used for midline catheter insertion (see later for a description of ultrasound guided cannulation). When properly placed and maintained, midline catheters can remain functional for weeks; e.g., in one clinical study, the average dwell time for midline catheters was 14 days (11). This same study also demonstrates that *vasopressors can be infused through midline catheters for as long as 7–8 days* without evidence of extravasation or limb compromise (11).

The extended dwell time of midline catheters have made them a popular choice when more than a few days of venous access is anticipated. In fact, midline catheters are now considered a safer alternative to the much longer “peripherally inserted central catheters” or PICCS (described in the next section), which have traditionally been favored for prolonged venous access. Comparative studies have shown that midline catheters have a lower incidence of catheter occlusions and catheter-related bloodstream infections than PICCs (12). Although there is a slightly higher risk of deep vein thrombosis (DVT) with midline catheters, the incidence of DVT with midline catheters is low (<5%), and the difference between midlines and PICCs is small (<2%) (13).

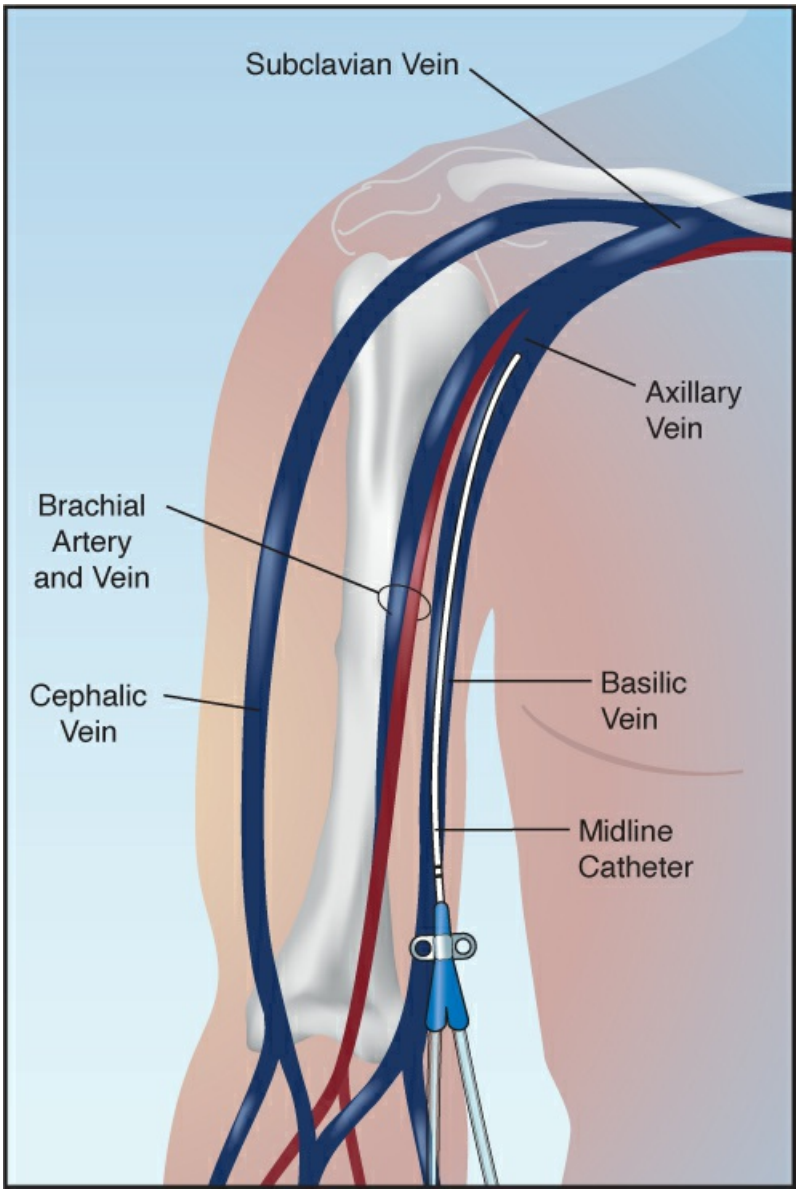


FIGURE 1.3 Illustration depicting the major veins un the upper arm, with a midline catheter in the basilic vein (the favored vein for midline catheter insertion).

TABLE 1.2 Comparative Features of Selected Peripheral Catheters

Catheter	Size	Length	Flow (L/hr)†
Short Peripheral Catheter[1]	18 ga	3 cm 1.2 in	6.0
Long Peripheral Catheter[2]	18 ga	8 cm 3.1 in	3.7
Double Lumen Midline Catheter[2]	Size: 5.5 French Lumens: 18 ga	15 cm 6 in	1.3/lumen

†All flow rates are for the gravity-driven flow of water from a height of 40 inches.

[1]Data from www.emupdates.com (accessed 5/17/2022).

[2]Data for Arrow® catheters, from www.teleflexvascular.com (accessed 5/17/ 2022). ga = gauge size, Fr = French

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size.

Midline catheters are available in lengths of 15 cm and 20 cm, and can have one to three infusion channels. The double-lumen catheter that is 15 cm in length is a popular choice, and some features of this catheter are included in Table 1.2. Note that the increased length of the midline catheter is accompanied by a decrease in flow capacity; in this case, the gravity-driven flow rate in each lumen of the midline catheter is only about 20% of the flow rate in the short peripheral catheter. Despite this flow decrement, the gravity-driven flow capacity in the midline catheter still exceeds the maximum flow provided by volumetric infusion pumps (i.e., 1 L/hr).

Central Catheters

As mentioned earlier, central vein catheters have a tip in one of the vena cavae. There are two types of central catheters, based on the location of the insertion site: *peripherally inserted central catheters* (PICCs) are inserted into one of the veins in the upper arm, while *centrally inserted central catheters* (commonly known as *central venous catheters*) are inserted into one of the major veins near the thoracic inlet, or in the groin.

Peripherally Inserted Central Catheters

Peripherally inserted central catheters (PICCs) are essentially a longer version of the midline catheters: i.e., they are inserted via the same veins as the midline catheters, but are long enough to be advanced into the superior vena cava. These catheters have been popularized as a safer alternative to central venous catheters when peripheral vein cannulation is problematic, or when more than one week of vascular access is anticipated (14). However, the emergence of midline catheters as a safer alternative to PICCs has led to a steady decline in the popularity of PICCs in hospitalized patients.

PICCs are available in lengths ranging from 40 cm to 55 cm, and can have one to three infusion channels. A popular choice is the double-lumen PICC that is 50 cm in length, and some features of this catheter are shown in Table 1.3. Note that the extended length of the PICC results in a marked decrease in flow capacity: e.g., the PICC in Table 1.3 is more than three times the length of the midline catheter in Table 1.2, and the resulting flow capacity is only 20% of that in the midline catheter. The compromised flow capacity in PICCs may explain the relatively high incidence of occlusions in these catheters (12).

Central Venous Catheters

Central venous catheters (CVCs) have played a prominent role in providing both secure and multifunctional venous access in critically ill patients. (The insertion of CVCs is described in detail in the next chapter.) These catheters are available in lengths of 15 cm, 20 cm, and 30 cm, and can have as many as 4 infusion channels. The most popular CVC is a triple-lumen catheter like the one shown in Figure 1.4. Note that the distal lumen has a larger bore (16 gauge) than the proximal or medial lumens (18 gauge); as a result, the flow capacity in the distal lumen is more than double that in the other lumens, as indicated in Table 1.3. For this reason, the distal lumen of a CVC is best suited for rapid volume infusions.

TABLE 1.3 Comparative Features of Selected Central Catheters

Catheter	Size	Length	Lumens	Flow (L/hr)†
Triple Lumen CVC	7 Fr	20 cm (8 in)	Distal (16 ga) Medial (18 ga) Proximal (18 ga)	2.28 0.91 0.99
Double Lumen PICC	5.5 Fr	50 cm (20 in)	Distal (18 ga) Proximal (18 ga)	0.26 0.27

†All flows rates are for the gravity-driven flow of water from a height of 40 inches.

Data for Arrow® catheters, from teleflexvascular.com (accessed 5/18/2022).

CVC = central venous catheter,

PICC = peripherally inserted central catheter, ga = gauge size, Fr = French size.

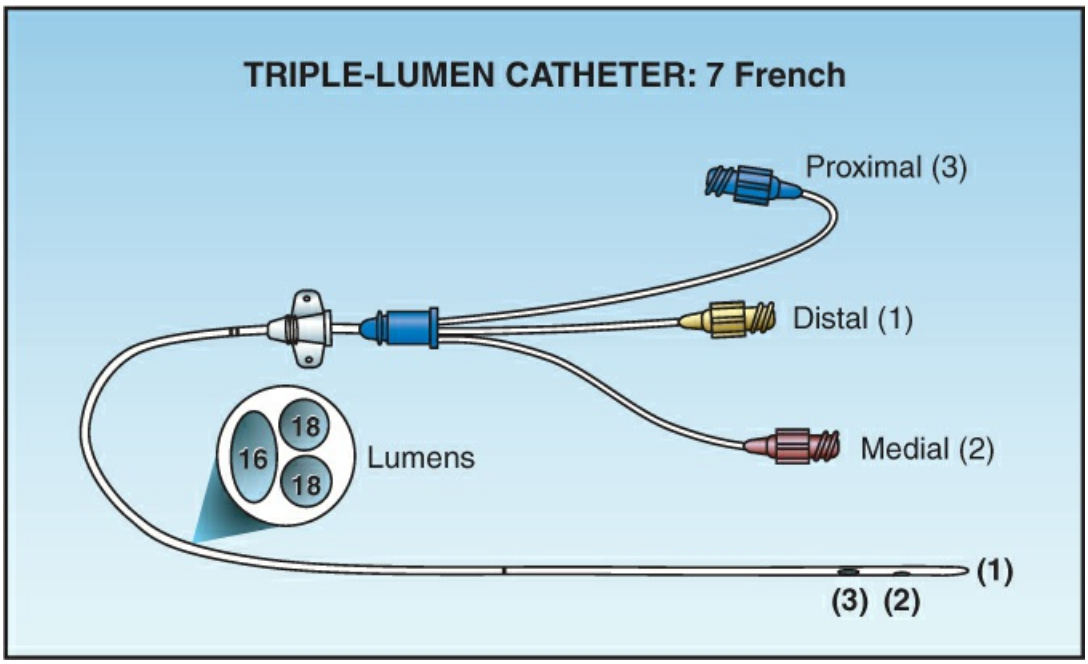


FIGURE 1.4 A triple-lumen central venous catheter, showing the gauge size of each lumen and the position of the outflow ports at the distal end of the catheter. This is currently the most popular design for central venous catheters.

The insertion of CVCs has been a staple of patient care in ICUs, but the popularity of CVCs is waning, as safer alternatives (i.e., PICCs and midline catheters) have emerged. (For more on the insertion of CVCs, see Chapter 2.)

The Guidewire

Cannulation of all but the most superficial, palpable veins is accomplished by advancing the catheter over a guidewire that is placed in the lumen of the vein. This technique of guidewire-assisted vascular cannulation, first introduced by a Swedish angiographer named Sven-Ibar Seldinger (and known as the *Seldinger technique*) (15), is illustrated in Figure 1.5. A small bore needle is used to probe for the target vessel. When

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the tip of the needle enters the vessel, a thin wire with a flexible tip is passed through the needle and into lumen of the blood vessel. (The flexible tip reduces the risk of en dothelial injury as the guidewire is advanced.) The needle is then withdrawn over the guidewire, and the catheter is advanced over the guidewire and into the lumen of the blood vessel. (In actual practice, a “dilator catheter” is first threaded over the guidewire to create a tract that facilitates catheter insertion.)

Summation

The advantages and disadvantages of general-purpose vascular catheters are summarized in Table 1.4. The following is a brief summary of how these catheters are used for patient care in the ICU. Short peripheral catheters are preferred for quick intravenous (IV) access at the outset, but in patients who spend more than a few days in the ICU, more stable IV access is needed. This has been traditionally supplied by CVCs, but over the past decades, PICCs began to replace CVCs for longer-term IV access. More recently, midline catheters have emerged as an acceptable choice for longer-term IV access, and they are replacing CVCs and PICCS for this purpose. However, PICCS continue to be popular for patients who require extended (several weeks) IV therapy. The major drawback of midline catheters and PICCs is the use of specially trained teams to insert the catheters; this limits the availability of these catheters.

Central venous catheters are the final (go-to) catheters for stable venous access, and are especially preferred in patients with life-threatening hemodynamic instability.

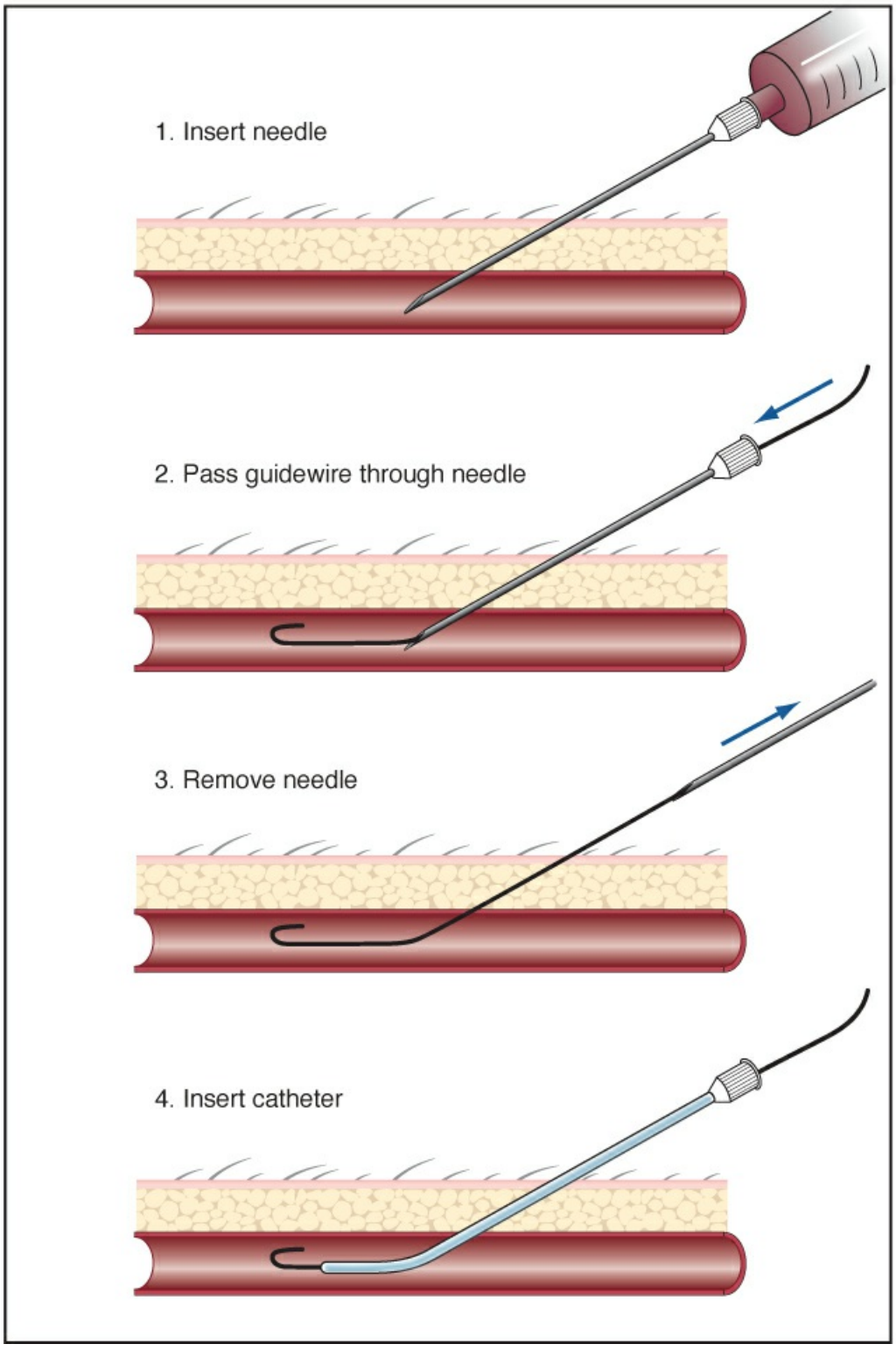


FIGURE 1.5 The steps involved in guidewire-assisted cannulation of blood vessels (the Seldinger technique).

TABLE 1.4 Summary of the General-Purpose Venous Catheters

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Type of Catheter	Advantages	Disadvantages
Peripheral Catheters		
Short Catheter	Rapid IV access Low risk of septicemia	Limited dwell time High failure rate
Long Catheter	Longer dwell time than short catheters	Can require ultrasound-guided insertion
Midline Catheter	Longest dwell time of peripheral catheters Multiple lumens Allows prolonged vasopressor infusions	Requires ultrasound-guided insertion and special training
Central Catheters		
Peripherally Inserted Catheter	Less risk, and greater patient acceptance, than central venous catheters	High occlusion rate Increased length can compromise infusion rate
Central Venous Catheter	Most versatile general-purpose catheter	Risk of serious complications

VASCULAR ULTRASOUND

The emergence of real-time ultrasound to guide vascular cannulation has added considerably to the success rate and safety of cannulating both central and peripheral veins (16,17). The following is a brief introduction to the methodology.

The Method

Vascular ultrasound uses linear array probes that emit high-frequency ultrasound waves (5–15 MHz); this produces high-resolution images, but limits the depth of tissue penetration (to about 9 cm). The waves that are reflected back to the probe (called *echoes*) are processed by a transducer that converts the ultrasound waves to gray scale images. Higher amplitude echoes produce brighter or whiter images, while lower amplitude echoes produce darker or blacker images. This methodology is known as B-mode (brightness-mode) ultrasound, and it produces two-dimensional, gray-scale images. Ultrasound waves pass readily through fluids, so blood vessels will have a dark gray or black interior on the ultrasound image.

Orientation of the Beam

The ultrasound beam can be aligned along the long axis of a blood vessel to produce a sagittal image of the vessel (long-axis view), or it can be oriented to transect a blood vessel, which produces a cross-sectional view of the vessel (short-axis view). This is demonstrated in Figure 1.6. Note that in the long-axis view, the probe needle advances along the plane of the ultrasound beam, and can be visualized along its entire path, while in the short-axis view, the probe needle does not meet the ultrasound beam until it reaches the target vessel, where it is visible only as a high-intensity dot on the ultrasound image.

Which View is Better?

Neither view is clearly superior to the other, and each has advantages and disadvantages. The advantage of the short axis view is greater ease in locating the target vein, and the ability to visualize both the vein and its affiliated artery (which reduces the risk of arterial puncture), while the disadvantage is the limited ability to view the probe needle. The long axis view allows visualization of the probe needle, guidewire, and catheter during the cannulation, but is unable to visualize an affiliated artery, and requires a vein that runs a relatively straight course.

At the present time, the short-axis view seems to be the preferred one for locating the target vessel and confirming puncture of the vessel with the probe needle. The long-axis view is preferred for confirming intraluminal placement of the guidewire and catheter (see next).

Protocol

The recommended protocol for ultrasound-guided vascular cannulation includes the following sequential steps (16): a) identify the target vein, b) verify puncture of the vein by the probe needle, and c) confirm intraluminal placement of the guidewire and catheter.

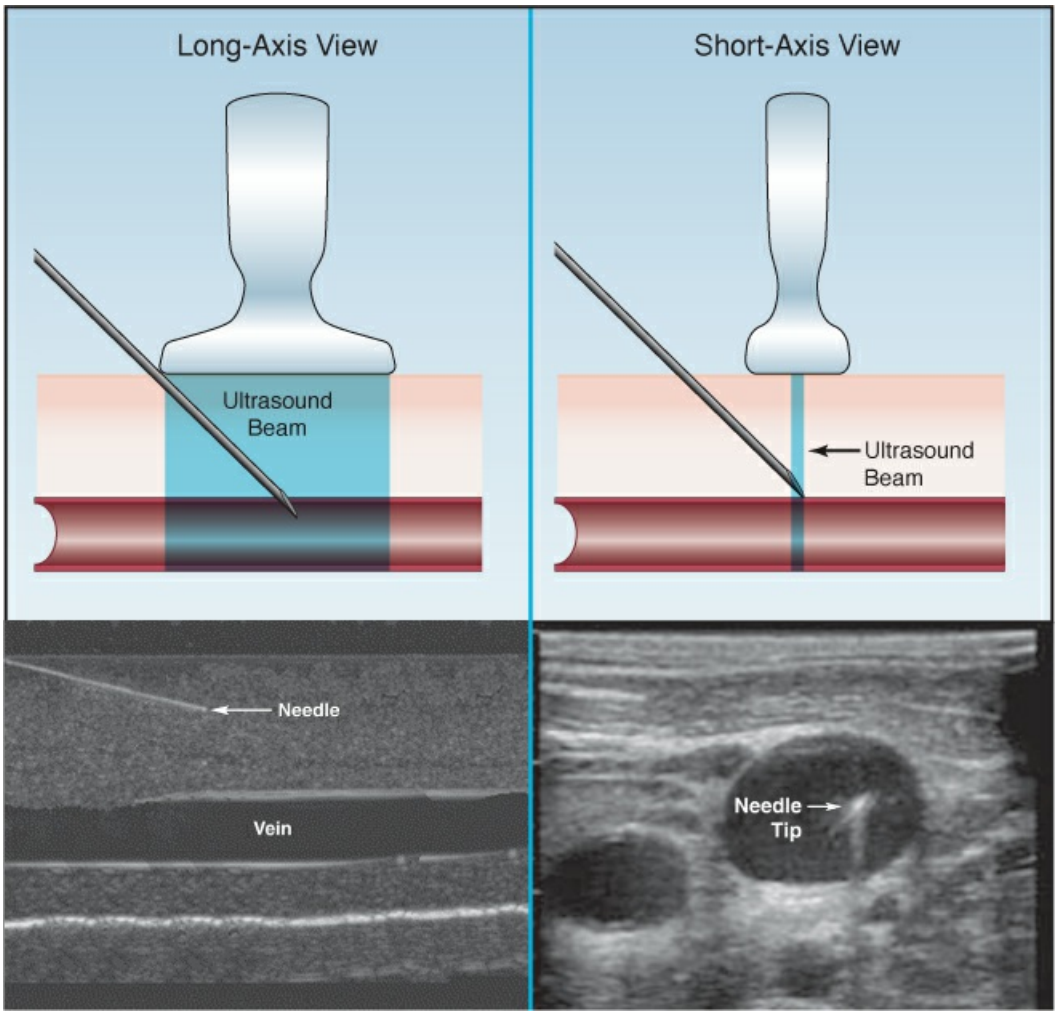


FIGURE 1.6 Orientation of the ultrasound beam in the long-axis and short-axis view, and how this influences the ability to visualize the probe needle. See text for further explanation.

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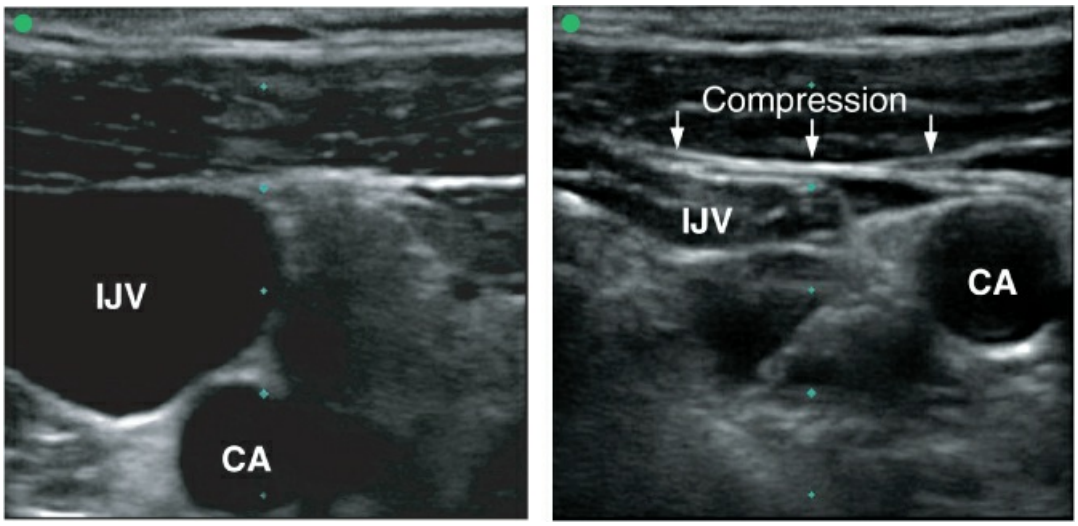


FIGURE 1.7 Compressibility for identifying veins. The image on the left is a short-axis view of the internal jugular vein (IJV) and carotid artery (CA) at the base of the neck, and the image on the right shows compression of the vein when downward pressure is applied to the overlying skin. The green dots mark the lateral side of each image.

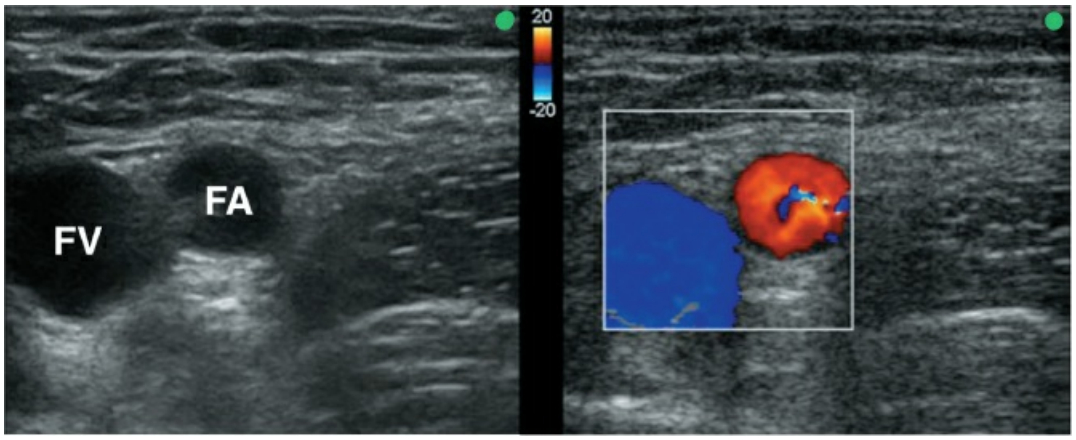


FIGURE 1.8 Color flow Doppler imaging for identifying arteries and veins. The image on the left is a short-axis view of the femoral artery (FA) and femoral vein (FV) in the groin, and the image on the right is a color Doppler image of the same blood vessels showing the artery in red and the vein in blue. The green dots mark the lateral side of each image.

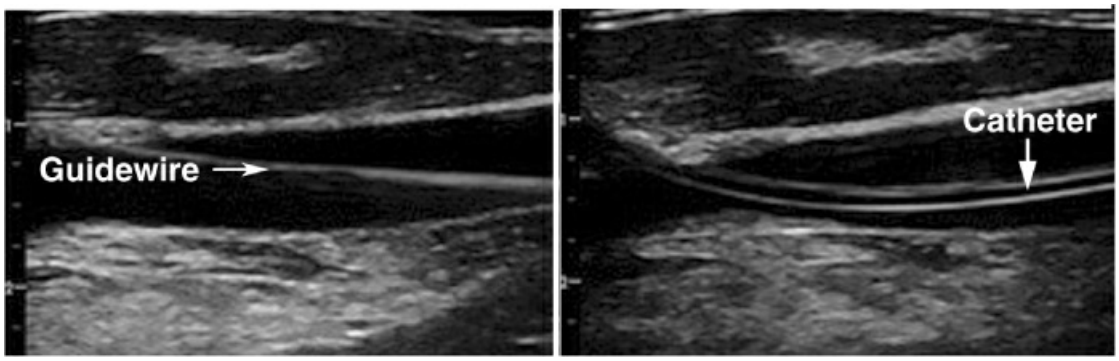


FIGURE 1.9 Long-axis view of the internal jugular vein showing intraluminal placement of the guidewire and catheter. Images from Reference 16.

Distinguishing Veins from Arteries

There are two methods for distinguishing arteries from veins using ultrasonography. The easiest (and most popular) method is to determine the compressibility of the blood vessel; i.e., when pressure is applied to the skin overlying a blood vessel, a vein will collapse much more readily than an artery. This is demonstrated in Figure 1.7, which shows an internal jugular vein that is compressed by pressing down on the overlying skin, while the nearby carotid artery is unaffected. (*Note:* One exception to this behavior is the presence of venous thrombosis, because a vein that is filled with a thrombus will not be compressible).

The other method for identifying arteries and veins is color Doppler imaging, where the frequency shift produced by the direction of blood flow (the Doppler effect) is converted into color images, and superimposed on the gray-scale ultrasound image (18). An example of this is shown in Figure 1.8, with the red color marking the femoral artery and the blue color marking the femoral vein. These colors are not specific for arteries and veins, but instead represent the direction of blood flow in relation to the ultrasound probe. (The color assignments are indicated on the color legend in Fig. 1.8.)

The Probe Needle

Advancing the probe needle to puncture the target vein is easily visualized in the long-axis view, as shown in Figure 1.6. In the short-axis view, the probe needle is advanced using short, stabbing thrusts to produce tissue displacement, which marks the path of the needle, and puncture of the target vein is verified by observing a high-intensity dot in the lumen of the vessel, as shown in Figure 1.6.

Confirming Catheter Placement

Veins tend to collapse when punctured, creating a tendency for probe needles to puncture the posterior wall of the vein. In one study, posterior wall perforation occurred in 40% of ultrasound-guided cannulations using the short-axis view, and 18% of cannulations using the long-axis view (19). Because of this risk (and the popularity of the short-axis view), it is important to confirm that the catheter has been placed in the lumen of the blood vessel. This begins by confirmating that the guidewire has been advanced into the vessel lumen, as shown on the left in Figure 1.9 (16). The catheter can then be advanced over the guidewire, and the intraluminal placement of the catheter is then confirmed, as shown on the right in Figure 1.9.

INTRAOSSEOUS VASCULAR ACCESS

The discovery that fluids could be infused into the marrow cavity of bones and reach the systemic circulation occurred in the mid-1930s. However, this observation gained little attention until the London bombings during World War II, when the poor lighting from the frequent citywide blackouts created difficulties in establishing venous access, and an enterprising surgeon named Hamilton Bailey began infusing fluids into the marrow cavity of the sternum to resuscitate bombing victims (20). Despite early enthusiasm, the intraosseous (IO) route was relegated to obscurity when plastic catheters were introduced for intravenous cannulation (in the 1950s). A rebirth of the IO route occurred in the 1980s, when it was adopted for pediatric resuscitation, and it has subsequently gained favor in adults as an alternative route for emergency

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vascular access.

Indications

The principal indication for IO access is the need for emergent vascular access when intravenous (IV) access is problematic, or is not immediately available. This scenario is most likely to occur in patients with cardiac arrest, major trauma, or circulatory shock; in each of these conditions, IO access has proven to be a viable alternative to IV access (21−23). One appealing feature of the IO route is the rigid structure of the medullary cavity and its drainage system, which (unlike veins) will not collapse in the setting of hypotension, hypovolemia, or cardiovascular collapse.

Acceptance of the IO route is demonstrated in the most recent guidelines on advanced life support from the American Heart Association, which recommends proceeding to IO access when an initial attempt at IV access is unsuccessful, or IV access is not feasible (21). Since about two-thirds of cases of cardiac arrest occur outside the hospital (24), the need for IO access occurs primarily in the field. As a result, prehospital personnel (emergency medical technicians and paramedics) have been the major focus of training for IO access, and the success rate in the field is as high as 97% (25). IO access in the field is also important for the early management of major trauma (22), including combat casualties (25), and IO access is included in the knowledge base for Advanced Trauma Life Support (ATLS) certification from the American College of Surgeons (26).

Contraindications

Contraindications to IO access include a fractured or previously entered bone (because of possible leakage of infused fluids), vascular injury in the target extremity, and burn injury, cellulitis, or osteomyelitis at the cannulation site. Osteopetrosis (abnormally dense bone) is also a contraindication (23), but is a rare condition.

Establishing IO Access

There are two popular sites for IO access in adults: the proximal tibia (just below the knee), and the proximal humerus (just below the shoulder). The proximal tibia has two advantages: a higher success rate (25,27), and a location that does not interfere with intubation or chest compressions. The proximal humerus has the advantage of a higher flow capacity (28), but the proximal tibia is the favored site in adults.

The Proximal Tibia Site

The access site in the proximal tibia is on a flat surface of bone just below the medial condyle (see Figure 1.10). The point of needle insertion is 3 cm below the inferior tip of the patella, and 2 cm medial to that point.

Inserting the Needle

Inserting the IO needle can be done manually, but a powered “needle driver” is preferred. A popular choice in recent years is a battery-powered device like the one shown in Figure 1.10 (28,29). There are three different IO needles that attach to this device: each is a 15 gauge needle, available in lengths of 15 mm (for children), 25 mm (for average-sized adults), and 45 mm (for large or obese adults). The appropriate length for an individual patient can be determined by first

advancing a probe needle until it hits the bone. (In awake patients, the probe needle should also be used to inject lidocaine locally, especially at the level of the periosteum, which is well endowed with pain fibers.) The IO needles have horizontal markers, and the measured distance from the skin to the periosteum should leave the horizontal marker closest to the hub of the needle at or slightly above the skin surface.

After appropriate skin antisepsis (sterile gloves are not required for inserting IO needles), the IO needle is advanced manually until it hits the bony surface of the tibia. The needle driver is then attached and, with the needle at an angle of 90° from the skin surface, the needle driver is engaged. This will spin the needle in a clockwise direction, and the needle is then advanced while applying downward pressure, until a sudden loss of resistance indicates entry into the medullary cavity. This should be verified by the aspiration of blood. This is not always possible, because bone marrow is viscous (like a jelly), which can prevent the aspiration of blood. Flushing the needle with 5–10 mL of saline can liquify enough marrow to correct this problem.

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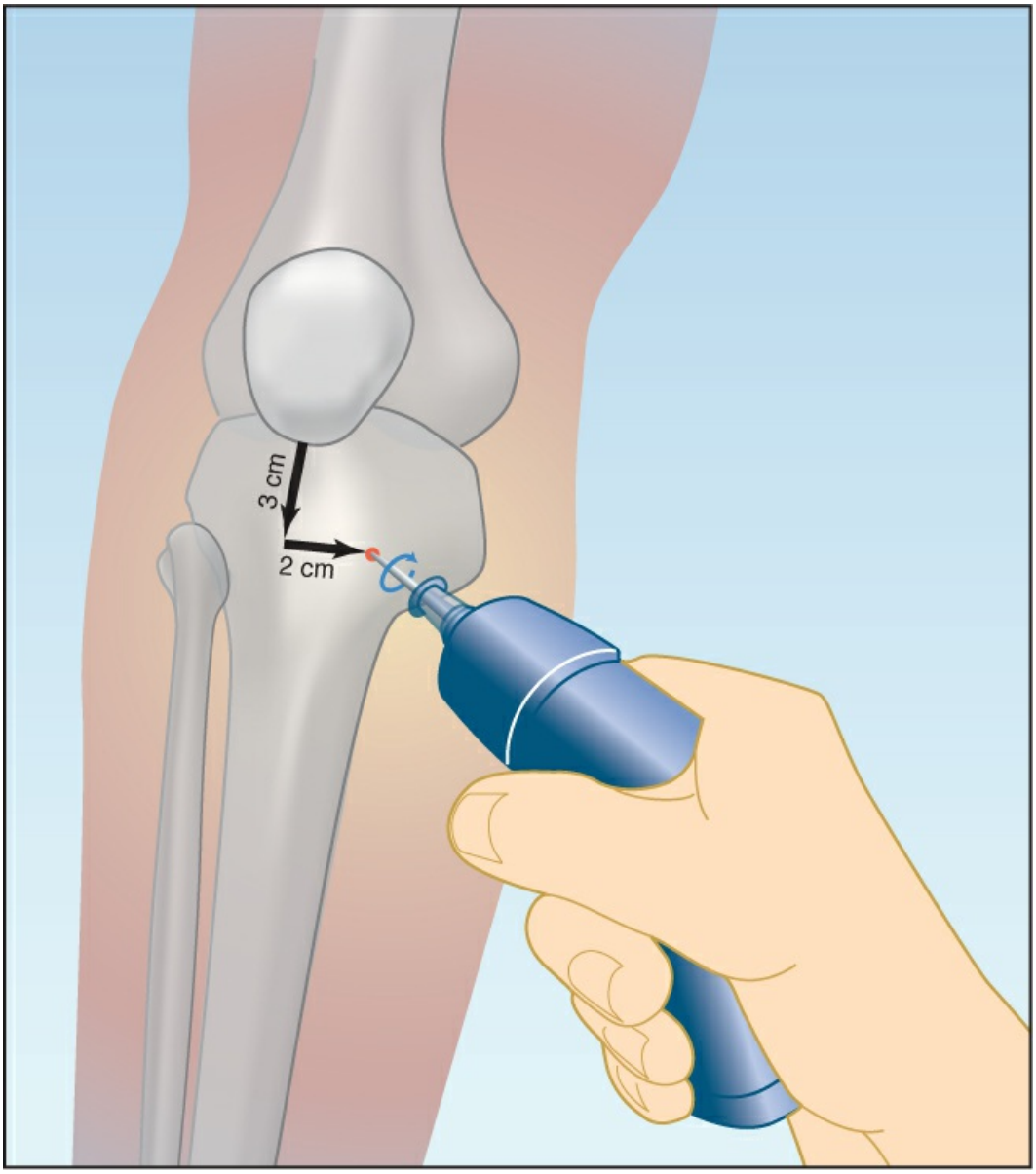


FIGURE 1.10 Illustration depicting the insertion of an intraosseous needle into the proximal femur, of the right leg and the surface measurements used to identify the insertion point. A battery powered needle driver is used to advance the needle into the marrow cavity. See text for further explanation.

Pre-Infusion Pain Control

The infusion of fluids into the marrow cavity is painful, presumably from stretching the periosteum. Therefore in awake patients, lidocaine injection into the medullary cavity is advised prior to infusing fluids. The recommended protocol is as follows (23,30):

. Use a 2% solution (20 mg/mL) of preservative-free lidocaine for intravenous use, and inject 40 mg (2 mL) into the marrow cavity over 2 minutes.

. Allow lidocaine to dwell for one minute, then flush with 10 mL of saline.

. Start the desired IO infusion. If pain relief is incomplete, readminister lidocaine at half the

initial dose (20 mg) over one minute.

The lidocaine effect may be lost after one hour, and additional analgesia may be necessary. Intraosseous fentanyl (in the usual IV doses) can be used for this purpose.

IO Infusions

Intraosseous infusions are relatively sluggish, with flow rates that are about 25% of peripheral vein infusions (22,23). This is due to the viscous nature of bone marrow, and the relatively high pressure in the marrow cavity (which averages about 30 mm Hg). To counter the sluggish flow rates, IO infusions are routinely pressurized (with infusion pumps or inflatable pressure bags). However, pressures of 300 mm Hg or even higher may be inadequate for replacing massive blood loss, and dual IO infusions have been advocated by some in this situation (22).

Intraosseous access is considered a temporary intervention, and IO infusions are not recommended for longer than 24 hours (26). (There is, however, no firm evidence to validate this recommendation.) When IO infusions are discontinued, the IO needle is removed by attaching a luer-lock syringe and slowly rotating the needle in a clockwise direction as it is pulled upward at a 90° angle with the skin surface. (Rocking the needle back and forth to facilitate removal is not advised, and will promote leakage from the bone.)

Complications

Complications of IO access are infrequent (possibly due to its limited lifespan), with several studies showing complication rates below 1% (22,23). Potential complications include skin abscesses, osteomyelitis, tibial fracture, compartment syndrome, and fat or marrow emboli (22,23,30,31).

A FINAL WORD

One of the rarest sights in any ICU is a patient with no intravenous access. While this may not be a revelation, it does indicate that a working knowledge of vascular access has universal relevance for patient care in the ICU. This chapter is a first step in acquiring that knowledge. The following points in the chapter deserve emphasis.

. Infusion rates are influenced by the dimension of a vascular catheter, not by the size of the cannulated vein.

. The radius (lumen size) of a catheter has a much greater influence on flow than the length of the catheter. For rapid infusions, a large bore catheter is desirable, and a short, large-bore catheter is optimal.

. Midline catheters (15 – 20 cm catheters inserted into one of the major veins above the antecubital fossa) have become a popular choice for extended venous access, and can be used for prolonged vasopressor infusions.

. Real-time ultrasonography has become an essential tool for cannulating veins that are not visible or palpable.

. The intraosseous (IO) route is recommended for emergency vascular access when intravenous access is problematic, or not immediately available. The proximal femur is a popular site for IO access because the procedure does not interfere with intubation or chest compressions.

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A L G r a w a n y

Chapter 2

Central Venous Access

Good doctors leave good tracks.

J. Willis Hurst, MD (*a*)

As a medical student in the early 1970s, I recall that vascular access was achieved almost exclusively by inserting narrow-bore needles into superficial veins in the arms (or wherever we could find them). The practice of inserting flexible plastic catheters into larger, more centrally placed veins was in its infancy (e.g., cannulation of the internal jugular vein was introduced only a few years earlier, in 1969) (1), and the procedure was often performed in the operating room, using a skin incision to introduce the (single-lumen) catheters, and forceps to advance the catheters through the subcutaneous tissues.

Now fast forward to modern times, and central venous access is a staple of patient care in ICUs, with multilumen catheters inserted percutaneously at the bedside, using real-time ultrasonography to guide the cannulation process (as described in Chapter 1). Central venous cannulation has become an essential skill for the care of critically ill patients, and this chapter will supplement the procedural skill (which must be acquired at the bedside) by presenting the relevant considerations involved in establishing central venous access.

CONSIDERATIONS IN PREVENTION

Central venous catheters have risks that far outweigh those of peripheral vein catheters (2), so the decision to insert a central venous catheter has implications for patient safety. There are, however, safer alternatives to central venous catheters in certain situations, as described next.

Shrinking Indications

The following are the traditional indications for inserting a central venous catheter.

. When peripheral venous access is difficult to obtain or maintain.

. For prolonged venous access (i.e., more than a few days).

. When multiple intravenous therapies are required (taking advantage of the multilumen

capabilities of central venous catheters).

. For the infusion of vasopressors, hypertonic fluids (including total parenteral nutrition), or vesicants (e.g., antineoplastic agents).

. For patients with life-threatening hemodynamic instability.

. For specialized interventions, such as acute hemodialysis, temporary transvenous pacing, or invasive hemodynamic monitoring.

It is now possible to avoid central venous access for many of the traditional indications, thanks to the emergence of midline catheters and peripherally inserted central catheters (PICCs). These catheters are described in Chapter 1. Both are inserted in the arm, just above the antecubital fossa (see Figure 1.3), and both can be used for prolonged venous access (especially PICCs, which can remain in place for months), and for infusions of vasopressors and hypertonic fluids (3). These catheters are also available with multiple lumens, for patients who require multiple intravenous therapies. Central venous access is still preferred for patients with life-threatening hemodynamic instability (circulatory shock), and it is mandatory for specialized interventions like acute hemodialysis. Thus, 4 of the 6 traditional indications for central venous access have alternative solutions.

Midline catheters and PICCs have fewer risks than central venous catheters (e.g., there is no risk of pneumothorax because the catheters are inserted in the arm) and they are more readily accepted by patients. Because of these advantages, the demand for central venous catheters is steadily declining (4), and midline catheters are becoming the preferred replacement (5).

Contraindications

There are no absolute contraindications to central venous cannulation, including the presence or severity of a coagulation disorder (6,7). There is a recommendation to correct a platelet count that is <20,000 x 10^6/L or an INR >3 prior to catheter insertion (7), but there is no evidence to support this recommendation.

Infection Control Measures

Infection control is an essential part of vascular cannulation, and the preventive measures recommended for central venous access are shown in Table 2.1 (8). When used together as a “bundle”, these measures have been effective in reducing the incidence of catheter-related bloodstream infections (9–11). The following is a brief description of these preventive measures.

TABLE 2.1 The Central Line Insertion Bundle

Components	Recommendations
Hand Hygiene	Use an alcohol-based, waterless hand rub or a soap and water handwash before and after: a) palpating the catheter insertion site, or b) inserting, replacing, or manipulating the catheter..
Barrier Precautions	Use maximal barrier precautions, including cap, mask, sterile gloves, sterile gown, and sterile full body drape, for catheter insertion or guidewire exchange.
Skin Antisepsis	Scrub the insertion site with 2% chlorhexidine -70% alcohol solution for 30 seconds, and allow to dry completely..

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Cannulation Site	When possible, avoid femoral vein cannulation.

From References 8–12.

Hand Hygiene

Proper hand hygiene is considered one of the most important, and most neglected, methods of infection control. Alcohol-based, waterless hand rubs are preferred, if available; otherwise, handwashing with soap (plain or antimicrobial soap) and water is acceptable (8,12). Hand hygiene should be performed before and after palpating catheter insertion sites, and before and after donning gloves to insert, replace, or manipulate catheters.

Skin Antisepsis

The catheter insertion site should be decontaminated just prior to cannulation, and the preferred antiseptic agent is chlorhexidine (usually in alcohol). This preference is based on clinical studies showing that chlorhexidine is superior to other antiseptic agents for limiting the risk of catheter-associated infections (13). The enhanced efficacy of chlorhexidine is attributed to its prolonged (≥6 hours) antimicrobial activity on the skin, which is maximized if allowed to air-dry on the skin (14). The drying time is typically 30 seconds for dry skin, and up to 2 minutes for moist skin (11).

Barriers

All vascular cannulation procedures, except those involving small peripheral veins, should be performed using full barrier precautions, which includes caps, masks, sterile gloves, sterile gowns, and a sterile drape from head to foot. (*Note*: The only barrier precaution advised for peripheral vein cannulation is the use of gloves, and nonsterile gloves are acceptable as long as the gloved hands do not touch the catheter) (8).

Site Selection

Cannulation of the femoral vein is considered the least desirable of the central venous access sites, primarily due to an increased risk of thrombosis with femoral vein catheters (15). When the central line bundle was introduced, there was also a higher risk of catheter-related septicemia with femoral catheters. However, *studies in more recent years have shown no increase in the risk of catheter-related septicemia with femoral catheters* (16). This is attributed to increased attention to infection control measures, and the current belief is that *the risk of catheter-related bloodstream infections is not related to the location of the insertion site, but rather to how the site is maintained* (in terms of infection control measures) (11).

INTERNAL JUGULAR VEIN

The most popular site for central venous access is the internal jugular vein at the base of the neck. The right side is preferred because the vessels run a relatively straight course, which reduces the risk of catheter misplacement.

Anatomy

The internal jugular vein (IJV) is located under the sternocleidomastoid muscle on either side of the neck, and it runs obliquely down the neck along a line drawn from the pinna of the ear to the sternoclavicular joint. In the lower neck region, the vein is typically located just anterior and lateral to the carotid artery (although anatomic relationships can vary), and both vessels run through the triangle created by the two heads of the sternocleidomastoid muscle (see Figure 2.1). At the base of the neck, the IJV and subclavian vein join to form the innominate vein, and the convergence of the right and left innominate veins forms the superior vena cava.

Cannulation Techniques

Positioning

Tilting the body so the head is below the horizontal plane (the Trendelenburg position) distends the IJV to facilitate cannulation, and increases venous pressure to reduce the risk of air embolism. In healthy subjects, a head-down body tilt to 15° is associated with a 20–25% increase in the diameter of the IJV, while greater degrees of tilt have no further effect (17). Thus, *only a limited (15°) body tilt is needed for IJV cannulation*. The head-down tilt is not necessary (and is usually not tolerated) in patients with venous congestion (e.g., from heart failure), and it is not advised in patients with increased intracranial pressure.

The head should be rotated slightly in the opposite direction to straighten the course of the vein. Excessive head rotation (beyond 40° from midline) pulls the IJV over the carotid artery, and this overlap increases the risk of carotid artery puncture (18). This risk is highlighted by evidence of posterior wall puncture in as many as 40% of ultrasound guided IJV cannulations (19).

Ultrasound Guidance

When real-time ultrasound is used to guide cannulation of the IJV, there is an increased success rate, fewer cannulation attempts, a shorter time to cannulation, and a reduced risk of carotid artery puncture (20). As a result, ultrasound guidance is a standard practice for catheterization of the IJV (21). (*Note:* The technique of ultrasound-guided vascular cannulation is described in Chapter 1.)

The IJV is well suited for ultrasound imaging because it is close to the skin, and there are no intervening structures to interfere with transmission of the ultrasound beam. A short-axis (cross-sectional) view of the IJV on the right side of the neck is shown in Figure 2.2. This image was obtained by placing the ultrasound probe at the apex of the triangle formed by the clavicular and sternal heads of the sternocleidomastoid muscle. (The probe should be oriented so it transects the muscle.) Note that the IJV is anterior and lateral to the carotid artery, and that there is some overlap of the vessels. This overlap creates the risk of carotid artery puncture, as explained earlier, and mandates that intraluminal placement of the catheter is confirmed. (See Figure 1.9.)

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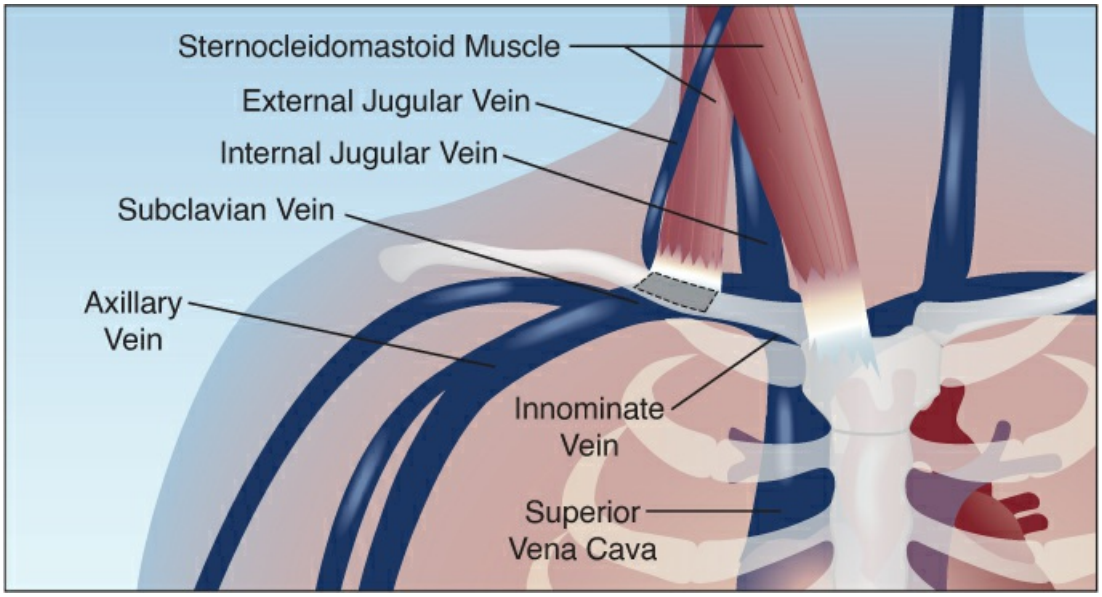


FIGURE 2.1 Anatomic relationships of the veins entering the thorax at the base of the neck.

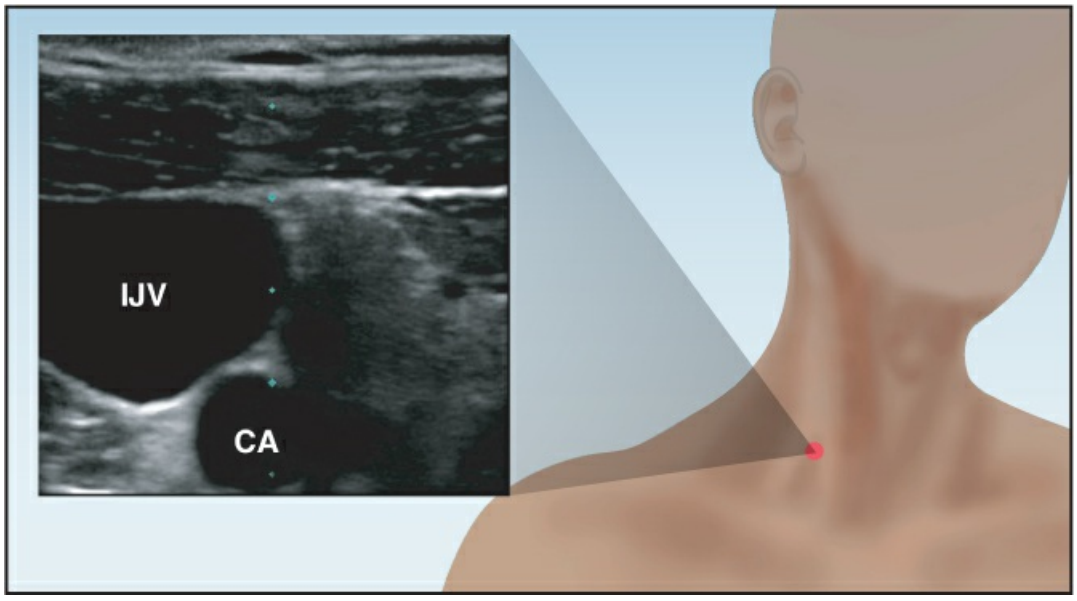


FIGURE 2.2 Short-axis view of the internal jugular vein (IJV) and carotid artery (CA), obtained by placing the ultrasound probe at the apex of the triangle formed by the two heads of the sternocleidomastoid muscle. The green dot marks the lateral side of the image. See text for further explanation.

Landmark Method

When ultrasound imaging is not available, cannulation of the IJV is guided by surface landmarks. There are two approaches using surface landmarks, as described next.

ANTERIOR APPROACH: For the anterior approach, the operator first identifies the triangular area at the base of the neck created by the separation of the two heads of the sternocleidomastoid muscle (see Figure 2.1). The IJV and carotid artery run through this triangle. The operator first locates the carotid artery pulse in this triangle; once the artery is located by palpation, it is gently

retracted toward the midline and away from the IJV. The probe needle is then inserted at the apex of the triangle (with bevel facing up) and the needle is advanced toward the ipsilateral nipple at a 45° angle from the skin. If the vein is not entered by a depth of 5 cm, the needle should be drawn back and advanced again in a more lateral direction.

POSTERIOR APPROACH: For the posterior approach, the insertion point for the probe needle is 1 cm above the point where the external jugular vein crosses over the lateral edge of the sternocleidomastoid muscle. The probe needle is inserted at this point (with the bevel at 3 o'clock) and then advanced along the underbelly of the muscle in a direction pointing to the suprasternal notch. The internal jugular vein should be encountered 5 to 6 cm from the insertion point.

Complications

Accidental puncture of the carotid artery is the most feared complication of IJV cannulation. In a review of six randomized controlled trials of IJV cannulation by experienced operators, the summed incidence of carotid artery puncture was 9.2% when anatomic landmarks were used, and 1.8% with ultrasound guidance (20). If the carotid artery is punctured by the small-bore probe needle, it is usually safe to remove the needle and compress the site for at least 5 minutes (double the compression time for patients with a coagulopathy). Insertion of a catheter into the carotid artery is more of a problem because removing the catheter can be fatal (22,23). *If confronted with accidental cannulation of the carotid artery, leave the catheter in place and consult a vascular surgeon or interventional radiologist.*

TABLE 2.2 Site-Specific Risks of Infection and Thrombosis

	IJV	SV	FV
Number of Catheters	845	843	844
Bloodstream Infection	0.5%	1.4%	1.2%
Symptomatic DVT	0.5%	0.9%	1.4%

Data from Reference 15. IJV = internal jugular vein, SV = subclavian vein, FV = femoral vein.

Other mechanical complications (e.g., hemo/pneumothorax) are less common, and have an aggregated incidence of 2% when anatomic landmarks are used, and <1% with ultrasound guidance (20). Finally, the risk of catheter-related septicemia with IJV catheters is slightly higher than the risk with subclavian catheters (15), but the difference is small (0.5% vs 1.4%, see Table 2.2), and the infectious risk at both sites is considered equivalent (11).

SUBCLAVIAN VEIN

The subclavian vein was once the favored site for central venous access, but the emergence of ultrasound guidance has eroded its popularity because of interference from the overlying clavicle.

A L G r a w a n y

Anatomy

The subclavian vein is a continuation of the axillary vein as it passes over the first rib (see Figure 2.1). It runs most of its course along the underside of the clavicle (sandwiched between the clavicle and the first rib), and at some points is only 5 mm above the apical pleura of the lungs. The underside of the vein sits on the anterior scalene muscle along with the phrenic nerve, which comes in contact with the vein along its posteroinferior side. Situated just deep to the vein, on the underside of the anterior scalene muscle, is the subclavian artery and brachial plexus. At the thoracic inlet, the subclavian vein meets the internal jugular vein to form the innominate vein. The subclavian vein is 3–4 cm in length, and the diameter is 7–12 mm in the supine position (24). The diameter of the vein does not vary with respiration (unlike the IJV), which is attributed to strong fascial attachments that fix the vein to surrounding structures and hold it open (24). These attachments may also prevent collapse of the vein with volume depletion.

Cannulation Techniques

Positioning

The head-down body tilt (Trendelenburg position) to 15° increases the diameter of the subclavian vein by about 10%, with no further effect from greater degrees of tilt (14). Despite this minimal effect, the increase in venous pressure in the Trendelenburg position (which reduces risk of air embolism) has justified the recommendation for the head-down body tilt during subclavian vein cannulation (25). Other popular maneuvers (aimed at bringing the subclavian vein closer to the clavicle), such as turning the head or placing a rolled towel under the shoulder, are not advised because they decrease the cross-sectional area of the vein (24,26).

Ultrasound Guidance

The use of real time ultrasound improves the success rate of subclavian vein cannulation (from 82% to 97%) and decreases the complication rate (from 30% to 11%) (27). However as mentioned earlier, ultrasound imaging is a challenge for the subclavian vein because of interference from the overlying clavicle. The vein can be visualized from above or below the clavicle, but the infraclavicular approach is the popular choice.

INFRACLAVICULAR APPROACH: This approach begins by identifying the clavicular head of the sternocleidomastoid muscle and its insertion on the clavicle; this marks the portion of the clavicle that overlies the subclavian vein (as shown in Figure 2.1). Orient the ultrasound probe so it transects the clavicle (with the orientation marker pointing cephalad), and place the probe just below the lower edge of the clavicle in this region. This should produce a short-axis view like the image in Figure 2.3, which includes the subclavian vein and artery, as well as the apical pleura. Note the proximity of the subclavian vein to the apical pleura, highlighting the risk of pneumothorax during the cannulation procedure.

Identifying the subclavian vein by compression may not be possible because of the overlying clavicle, and color Doppler imaging may be necessary (see Figure 1.8). At the end of the cannulation procedure, the pleural line should be inspected for evidence of *lung sliding* to rule out iatrogenic pneumothorax (see later).

Landmark Method

To cannulate the subclavian vein without ultrasound, first mark the region of the clavicle that overlies the subclavian vein, as shown in Figure 2.1. The vein can be entered from above or below the clavicle in this region, but the infraclavicular approach is the popular route.

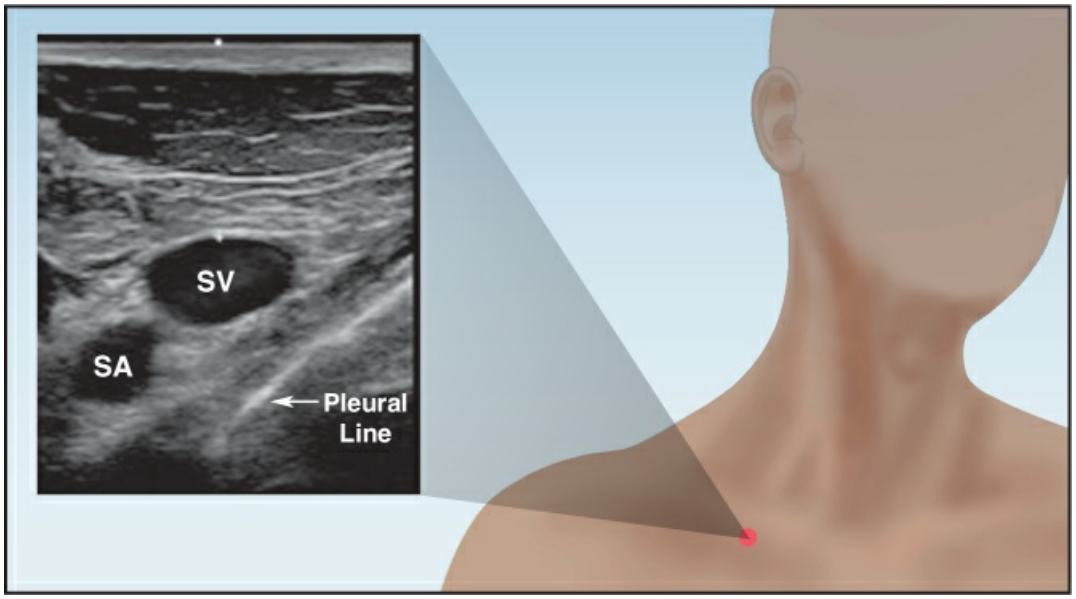


FIGURE 2.3 Infraclavicular, short-axis view of the subclavian vein (SV), subclavian artery (SA), and apical pleura, obtained by placing the ultrasound probe just below the clavicle in the region where it overlies the subclavian vein. See text for further explanation. Ultrasound image from Reference 25.

INFRACLAVICULAR APPROACH: At the lateral edge of the marked area, insert the probe needle (with the bevel at 12 o'clock) just below the clavicle, and advance the needle along the underside of the clavicle, in the direction of the suprasternal notch. The needle should enter the subclavian vein within a few centimeters. When the vein is punctured, turn the bevel to 3 o'clock to help direct the guidewire into the superior vena cava. It is important to keep the needle on the underside of the clavicle to avoid puncturing the subclavian artery (which lies deep to the subclavian vein) and the apical pleura.

In obese patients, the subclavian vein can be more deeply situated, and a deeper trajectory for the probe needle may be needed. This creates the risk for puncture of the subclavian artery or apical pleura. *In morbidly obese patients, the depth of the subclavian vein can exceed the reach of the probe needle* (26). For these reasons, the landmark approach to subclavian vein cannulation should be avoided, if possible, in morbidly obese patients.

Complications

The acute complications of subclavian vein cannulation (using ultrasound to landmark rates) include puncture of the subclavian artery (1% to 6%), pneumothorax (1% to 4%), brachial plexus injury (0% to 3%), phrenic nerve injury (0% to 2%), and catheter malposition (8% to 9%) (27). Complications associated with indwelling catheters include septicemia, thrombosis, and subclavian vein stenosis. The latter complication appears days or months after catheter removal,

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and has a reported incidence of 15–50% (28). The risk of stenosis is the principal reason to *avoid cannulation of the subclavian vein in patients who might require long-term hemodialysis access in the ipsilateral arm.*

FEMORAL VEIN

The femoral vein is considered the least desirable site for central venous access, although the unfavorable reputation is mostly undeserved (see later).

Anatomy

The femoral vein is a continuation of the long saphenous vein in the groin, and is the main conduit for venous drainage of the legs. It is located in the femoral triangle along with the femoral artery and nerve, as shown in Figure 2.4. The superior border of the femoral triangle is formed by the inguinal ligament, which runs from the anterior superior iliac spine to the pubic symphysis, just beneath the inguinal crease on the skin. At the level of the inguinal ligament (crease), the femoral vein lies just medial to the femoral artery, and is only a few centimeters from the skin. The vein is easier to locate and cannulate when the leg is placed in abduction.

Cannulation Techniques

Real-time ultrasound is recommended for femoral vein cannulation (21), although there is little evidence of benefit with ultrasound at this site (29).

Positioning

Elevation of the upper body to 15° above horizontal (the ”reverse” Trendelenburg position) can increase the cross-sectional area of the femoral vein by about 50% (30), so mild upper body elevation should be advantageous, especially for the landmark approach. Placing the leg in abduction is also recommended to facilitate the cannulation procedure.

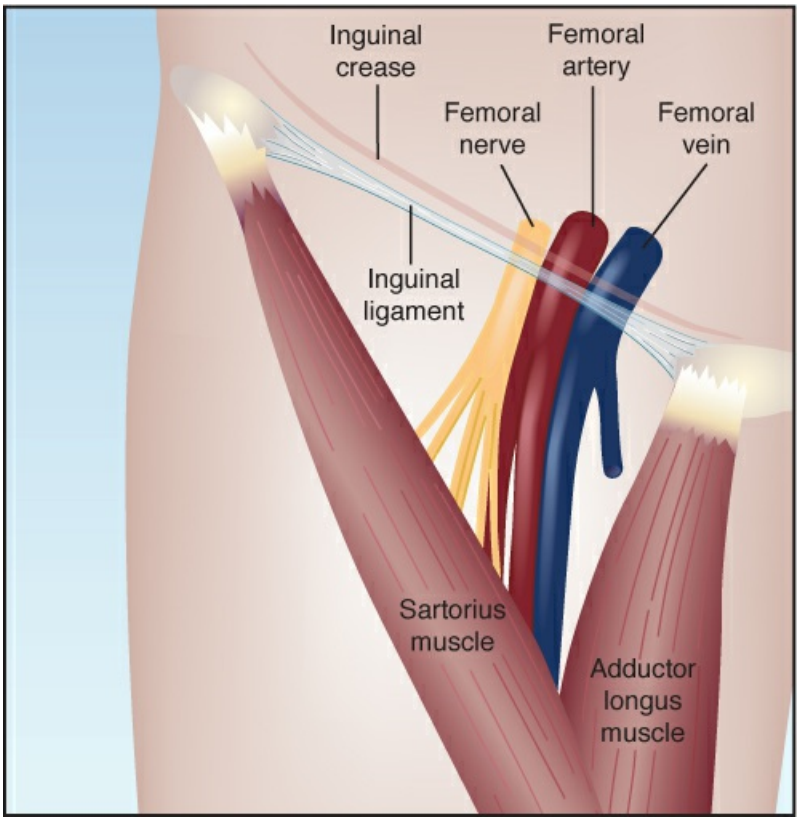


FIGURE 2.4 Anatomy of the femoral triangle.

Ultrasound Guidance

Ultrasound visualization of the femoral artery and vein is possible by placing the ultrasound probe over the femoral artery pulse, which is typically located just below and medial to the midpoint of the inguinal crease. A short-axis view of the femoral artery and femoral vein in this location is shown in Figure 1.8. Note that the vein lies a little deeper than the artery.

Landmark Method

To cannulate the femoral vein without ultrasound imaging, begin by locating the femoral artery pulse just below the inguinal crease. Then insert the probe needle (with the bevel at 12 o'clock) 1–2 cm medial to the pulse, and the vein should be entered at a depth of 2 to 4 cm from the skin. If the femoral artery pulse is not palpable, draw an imaginary line from the anterior superior iliac crest to the pubic tubercle, and divide the line into three equal segments. The femoral artery should be just underneath the junction between the middle and medial segments, and the femoral vein should be 1–2 cm medial to this point. This approach has a reported success rate of >90% (31).

Complications

The principal concerns at the femoral site are thrombosis and septicemia, which are considered enough of a risk to discourage femoral vein cannulation (as indicated in Table 2.1). However, study results like those in Table 2.2 (15) are not as foreboding as advertised. These results (from a multicenter study that included 2,532 central venous catheters) show a remarkably low rate of

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infectious and thrombotic complications at all sites. The risk of catheter-related septicemia at the femoral site was actually lower than the risk at the internal jugular vein site (the favored site for central venous access). The femoral site did have the highest risk of symptomatic deep vein thrombosis (DVT), but the overall risk was minor (1.5%). Results like these are supported by other studies (32), and they demonstrate the safety of the femoral vein site for central venous access.

IMMEDIATE CONCERNS

Venous Air Embolism

Air entry into the venous circulation is an uncommon but potentially lethal complication of central venous cannulation (33).

Pathophysiology

Pressure gradients that favor the movement of air into the venous circulation are created by the negative intrathoracic pressure generated during spontaneous breathing. A pressure gradient of only 5 mm Hg across a 14 gauge catheter (internal diameter = 1.8 mm) can entrain air at a rate of 100 mL per second, and this is enough to produce a fatal venous air embolism (34).

The impact of air entry into the venous circulation is determined by the volume of air and the rate of entry. The outcome can be fatal when air entry reaches 200–300 mL (3–5 mL/kg) over a few seconds (34). Entrained air can produce an air lock in the right ventricle, leading to acute right heart failure and cardiogenic shock, while air reaching the pulmonary circulation can produce leaky-capillary pulmonary edema (33,34). Finally, air can pass through a patent foramen ovale and produce an acute embolic stroke.

Prevention

The standard preventive measure for air embolism is the head-down body tilt (the Trendelenburg position) to increase venous pressure during cannulation of the internal jugular and subclavian veins. Elevation of the upper body (the reverse Trendelenburg position) is not necessary as a preventive measure during femoral vein cannulation because femoral vein catheters do not enter the thorax and thus are not exposed to negative pressures.

Clinical Presentation

Venous air entry can be clinically silent (33). In symptomatic cases. the earliest manifestation is sudden onset of dyspnea, which may be accompanied by a distressing cough. This can progress rapidly to acute respiratory failure and circulatory shock. In the most advanced cases, the mixing of air and blood in the right ventricle can produce a splashing auscultatory sound called a ‘mill wheel’ murmur (35).

Venous air embolism is usually a clinical diagnosis. Transesophageal echocardiography is considered the most sensitive method of detecting air in the right heart (capable of detecting as little as 0.02 mL/kg) (33), but this procedure is often not readily available. Precordial Doppler ultrasound can be useful (air in the cardiac chambers produces a high-pitched sound), but the Doppler signal can lack specificity (33).

Management

The following measures are recommended for the management of venous air embolism (33,34), although their efficacy is unproven.

. If air entrainment is suspected through an indwelling catheter, attach a syringe to the catheter and attempt to aspirate air from the bloodstream.

. Place the patient on 100% oxygen (to promote the movement of nitrogen out of the air bubbles in the bloodstream, and thereby decrease the volume of entrained air.

. Place the patient in the left lateral decubitus position (to move an air pocket that is blocking the outflow of the right ventricle).

. For patients with cardiovascular collapse, consider extracorporeal support.

Pneumothorax

Pneumothorax is a feared but infrequent complication of central venous access, and most cases are associated with subclavian vein cannulation (where the incidence is 1% using ultrasound and 4% otherwise) (27). The detection of post-procedure pneumothoraces can be problematic, as explained next.

Portable Chest Radiography

The portable chest x-ray has been the standard method for detecting pneumothorax after central venous cannulation, but clinical studies have shown that *portable chest x-rays fail to detect as many as 50% of pneumothoraces in critically ill patients* (36,37). This lack of sensitivity is attributed to the supine position (which is the position of most patients in the ICU during portable chest radiography), because pleural air does not collect at the apex of the lungs in the supine position, but instead collects in the anterior region of the pleural cavity (which can be close to the base of the lung) (38,39). Pleural air in this location will be in front of the lungs, and can be missed on a portable chest x-ray because of the lung markings behind the pleural air. An example of an anterior pneumothorax that is not apparent on a supine chest x-ray is shown in Figure 2.5.

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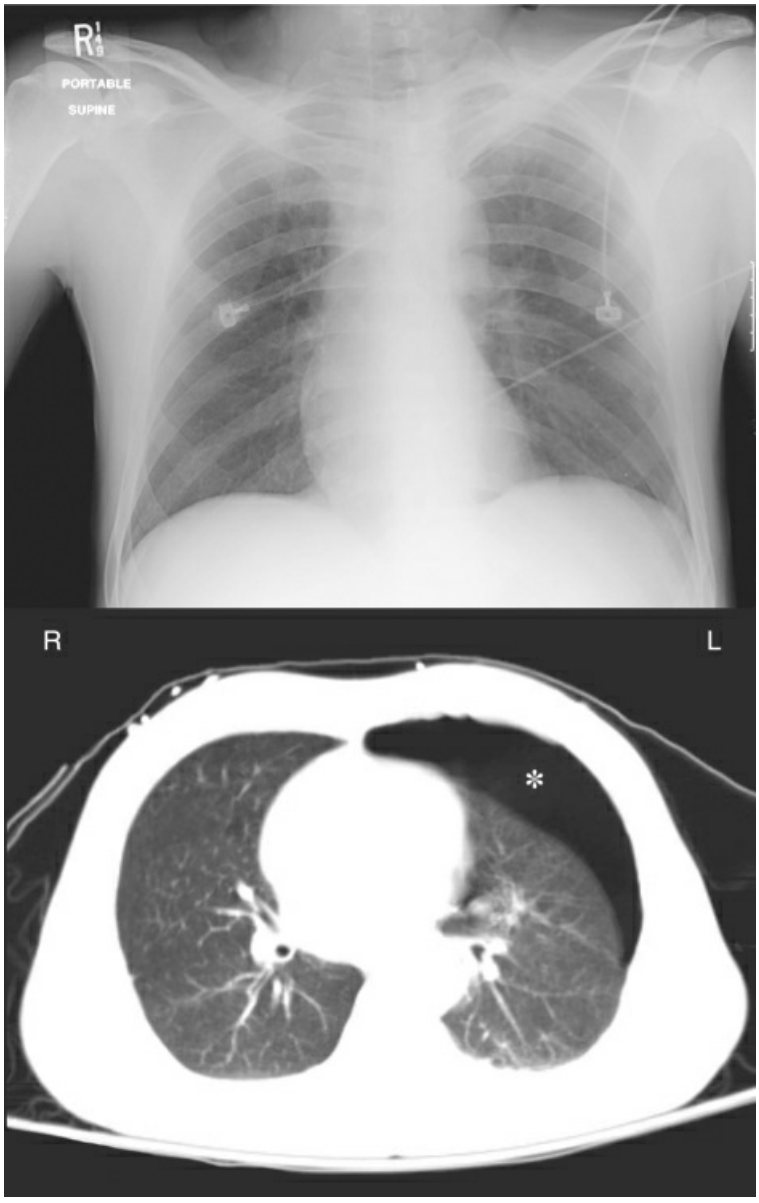


FIGURE 2.5 Portable chest x-ray and CT image of the chest in a young male with blunt chest trauma. The chest film is unrevealing, while the CT image shows an anterior pneumothorax (indicated by the asterisk).

Ultrasound

The pleura can be visualized on ultrasound imaging using a high-frequency, linear array probe (the same probe used for vascular ultrasound) that is placed across the intercostal spaces. The normal movement of the pleural surfaces creates a shimmering effect on the pleural image that is known as "lung sliding" (40). The absence of lung sliding then suggests the presence of a pneumothorax. Other conditions (e.g., blebs, pleurodesis) can be accompanied by the absence of lung sliding, so this sign is not pathognomonic of pneumothorax.

Several clinical studies have shown that ultrasound has a higher sensitivity than portable chest x-rays for the detection of pneumothoraces (36,37), including those associated with central venous cannulation (41). As a result, ultrasound has been recommended as a replacement for

chest radiography after central venous cannulation.

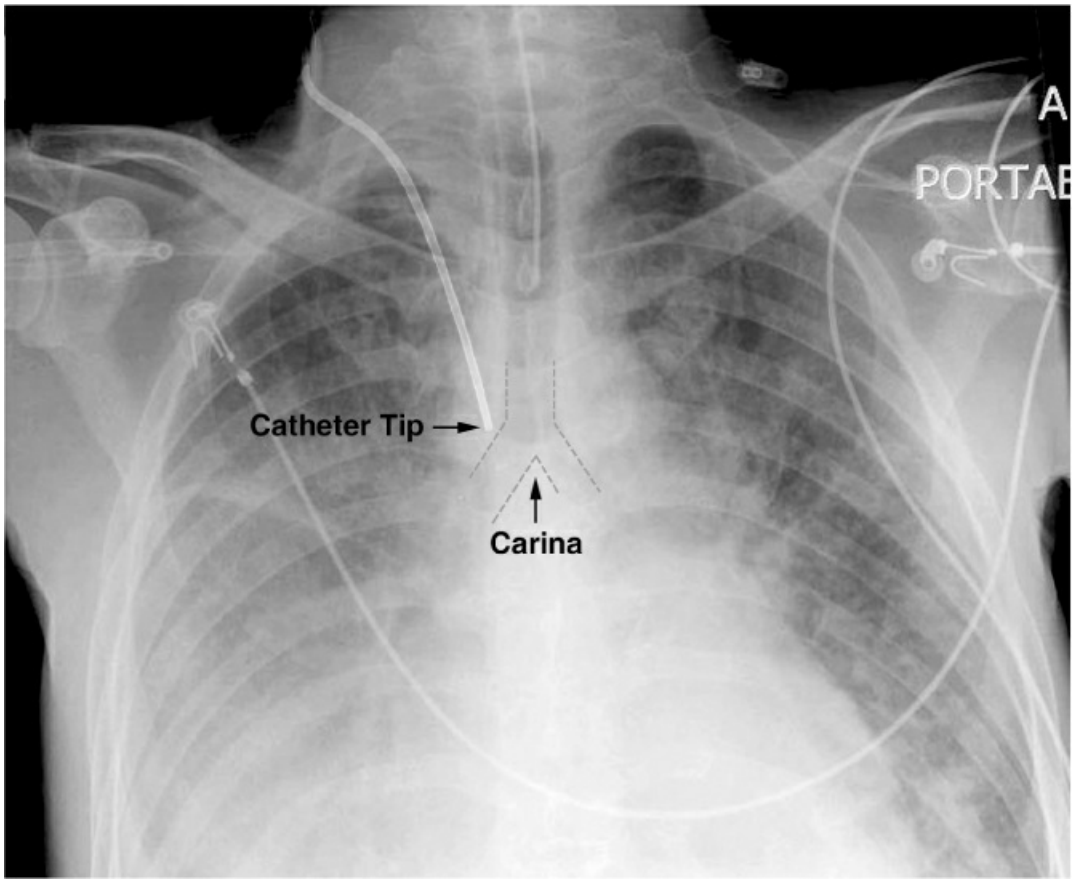


FIGURE 2.6 Portable chest x-ray showing the proper placement of a central venous catheter. Note that the tip of the catheter is at the level of the carina (i.e., the tracheal bifurcation, highlighted by the dotted lines), which lies just above the junction between the superior vena cava and right atrium. Image digitally enhanced

Catheter Position

Catheters inserted via the internal jugular and subclavian veins should be in the superior vena cava, with the tip 1–2 cm above the right atrium. Misplacement of catheters is reported in 15–18% of cannulations (41), mostly those involving the subclavian vein.

Chest Radiography

The standard practice is to evaluate catheter placement with a portable chest x-ray, and the one in Figure 2.6 shows a catheter that is appropriately placed. Note that the catheter follows a straight course down the mediastinum, and the tip of the catheter is just above the carina (i.e., the tracheal bifurcation). The carina is located just above the junction between the superior vena cava and the right atrium, so a catheter tip that is at the level of the carina, or slightly above it, is in the appropriate position. The carina is thus a useful landmark for evaluating catheter position (42).

Ultrasound

Ultrasound has been recommended as a replacement for chest radiography to evaluate catheter

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placement. There are two aspects of the ultrasound examination. The first pertains to subclavian vein catheters, and involves imaging the internal jugular vein on both sides to identify cephalad misplacement of the catheter (which must be corrected). This can be done during the catheterization procedure to save time in repositioning the catheter.

The second aspect of the ultrasound exam uses a phased array transducer in the subcostal window to visualize the catheter in the right atrium. If the catheter is not in the right atrium, then a "bubble study" is performed. This involves the rapid injection of 10 ml of saline through the distal port of the catheter, and observing for the appearance of microbubbles in the right atrium. A positive test confirms that the catheter is in the venous system, while the appearance of the bubbles within 2 seconds of injection is evidence that the catheter tip is in the superior vena cava (43).

Catheter Tip in Right Atrium

Catheters that have been advanced into the right atrium have traditionally been repositioned because of the perceived risk of right atrial perforation and cardiac tamponade. However, this practice is being questioned because of the rarity of this complication (44). In one study that included 2,348 patients with a catheter tip that remained in the right atrium, there were no cases of cardiac perforation or troublesome cardiac arrythmias (45). As a result of studies like this, the practice of repositioning catheters that enter the right atrium is being abandoned.

A FINAL WORD

The following points related to central venous access deserve emphasis.

. The indications for central venous access are shrinking, thanks to the emergence of safer alternatives like midline catheters and peripherally inserted central catheters (PICCs).

. The use of real-time ultrasound improves the success rate, and reduces the complication rate, at all sites of central venous access.

. The perception that femoral vein catheters are particularly risky is not supported by clinical studies (see Table 2.2).

. Ultrasound is superior to portable chest radiography for detecting post-insertion pneumothoraces, and is a suitable alternative to chest radiography for evaluating catheter position.

. Catheters that have been advanced into the right atrium pose no great danger of cardiac perforation, and can be left in place.

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Chapter 3

The Indwelling Vascular Catheter

See, and then reason and compare and control. But see first.

Sir William Osler (*a*)

The ubiquitous presence of indwelling vascular catheters requires attention to their maintenance and vigilance for their troublesome complications. This chapter begins by describing the routine care and maintenance of indwelling catheters, including how to restore patency in an occluded catheter, and then focuses on the major complications of indwelling catheters: i.e., thrombosis, vascular perforation, and catheter-related infections.

ROUTINE CATHETER CARE

The considerations involved in routine catheter care are summarized in Table 3.1.

Protective Dressings

Catheter insertion sites should be covered with a sterile dressing for the life of the catheter. There are three types of dressings (1–3):

. Sterile gauze pads, which are secured to the skin with hypo-allergenic tape.

. Transparent polyurethane dressings, which are permeable to O_2, CO_2, and water vapor, but trap moisture that collects on the underlying skin. These dressings have a skin adhesive that produces a seal around the catheter insertion site, and they are commonly known as "occlusive" dressings.

. Chlorhexidine gluconate-impregnated sponge dressings, which can be placed over the catheter insertion site, and then covered with the transparent dressing.

The transparent occlusive dressings are favored because they allow routine inspection of the catheter insertion site, and also provide a seal that isolates the insertion site. However, the skin adhesive on these dressings does not work well when the skin is wet, so gauze dressings are preferred when the skin is difficult to keep dry. The chlorhexidine-impregnated dressings are typically available as a small sponge (e.g., Biopatch®) that is included in many central venous

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catheter insertion sets.

Sterile gauze dressings and occlusive dressings are considered equivalent in preventing catheter colonization and infection (1–3). Chlorhexidine-impregnated dressings provide the best protection against catheter infections (1,2), but they are currently recommended only in settings where catheter-related infections are problematic (4).

TABLE 3.1 Recommendations for Routine Catheter Care

Task	Recommendations
Dressing Changes	Adhesive transparent dressings should be changed at least every 7 days, or (1) when moisture or blood accumulates under the dressing, or (2) the adhesive seal is disrupted. Gauze dressings should be changed at least every 48 hrs, or when the dressing is visibly soiled.
Flushing Catheters	Preservative-free 0.9% saline is preferred to heparinized saline for flushing venous catheters..
Replacing Catheters	Routine or scheduled replacement of catheters is not necessary, and can create complications.

Dressing Changes

The occlusive transparent dressings should be changed at least every 7 days, or when the adhesive seal is disrupted (2,3). In addition, *occlusive dressings can paradoxically promote colonization and infection when moisture or blood accumulates under the dressing* (5), and evidence of such should prompt a dressing change. Gauze dressings should be changed at least every 48 hours, or when the dressing is no longer dry (2,3).

Replacing Catheters

Routine or scheduled replacement of catheters is not advised (3) because it does not reduce the incidence of catheter-related infections (6), and actually increases the risk of catheter-associated complications (both mechanical and infectious) (7,8).

Peripheral Vein Catheters

Peripheral vein catheters are replaced when there is evidence of: (a) localized phlebitis (i.e., painful erythema around the insertion site, which can extend centrally along the course of the vein), (b) extravasation of the infusate, or (c) catheter obstruction that cannot be relieved (see later) (2). The phlebitis is typically the result of mechanical or chemical irritation of the target vessel, and does not represent infection (unless there is purulence emanating from the puncture site).

Central Venous Catheters

Central venous catheters are replaced when there is evidence of infection (either purulence at the insertion site or confirmed catheter-related septicemia). Catheter replacement is not necessary when there is erythema at the catheter insertion site without other signs of infection, since erythema alone is not predictive of catheter-related infection (9).

Flushing Catheters

Vascular catheters that are used intermittently are at risk of developing an occlusive thrombosis when not in use. To prevent this, the catheters are flushed at regular intervals, and the flush solution is left in the lumen of the catheter when it is not in use (known as "locking" the catheter). Flush solutions have traditionally contained heparin (in concentrations ranging from 10 to 1,000 units/mL, in isotonic saline), but these solutions are concerning because of the risk of heparin-induced thrombocytopenia (see Chapter 13). Isotonic saline has proven as effective as heparinized saline as a flushing and locking solution (10), and *preservative-free saline is now recommended as the preferred solution for flushing and locking venous catheters* (11). However, this is not the case for arterial catheters (12), where 1.4% sodium citrate is a suitable alternative to heparinized saline for maintaining catheter patency (13).

Occluded Catheters

Occlusion of indwelling vascular catheters is common, and can be the result of thrombus formation (60% of cases) mechanical factors (e.g., constriction from a suture), catheter malposition, or insoluble precipitates from drugs or minerals in the infusate (14). The occlusion can be partial or complete, and can appear as sluggish flow, loss of flow, or frequent triggering of alarms on infusion pumps (indicating excessive infusion pressures). Aspiration of blood through the catheter becomes difficult, or is not possible. (*Note:* Small-barrel syringes are preferred for aspiration through a catheter because they generate smaller negative pressures, and are less likely to collapse the lumen of the catheter.)

Restoring Patency

Short, peripheral vein catheters that become occluded are typically replaced, but for long catheters that are more difficult to replace (e.g., central venous catheters), every effort should be made to restore patency. Advancing a guidewire through the catheter to restore patency is not advised, since this can dislodge an obstructing mass and create an embolus. Chemical dissolution of the obstructing mass (described next) is the standard method of restoring patency.

THROMBOTIC OCCLUSION: Since thrombosis is the most common cause of catheter occlusion, the initial attempt to restore patency should involve the local instillation of a thrombolytic agent. Alteplase (recombinant tissue plasminogen activator) is currently the favored thrombolytic agent for restoring catheter patency; the regimen shown in Table 3.2 can restore patency in 80–90% of occluded catheters (15,16). There are 11 reports of angioedema associated with this regimen (17), and no reports of abnormal bleeding (16).

TABLE 3.2 Protocol for Declotting Central Venous Catheters

Drug	Alteplase (recombinant tissue plasminogen activator)
Preparation	Cathflo Activase (Genentech, Inc.) available as a powder in 2 mg vials. Add 2 mL sterile water to each vial for a drug concentration of 1 mg/mL.
Regimen	1. Instill 2 mL (2 mg) of drug solution into the occluded catheter and cap the hub of the catheter. 2. Wait 30 minutes and attempt to withdraw blood from the catheter.

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	3. If the occlusion persists, wait another 90 minutes (total dwell time = 120 min) and attempt to withdraw blood from the catheter. 4. If the occlusion persists, prepare a second dose of alteplase (2 mg) and repeat steps 1–3. 5. If patency is restored, withdraw 5 mL of blood through the catheter to remove any residual clot. 6. If this regimen does not restore patency, consider the non-thrombotic sources of obstruction in Table 3.3.

From Reference 16.

TABLE 3.3 Non-Thrombotic Catheter Occlusion

Etiology	Treatment	Regimen
Alkaline Drug Precipitates		1. Instill 1 mL or catheter lumen volume. 2. Dwell 60 minutes 3. Attempt flushing with 10–20 mL 0.9% saline. 4. Repeat if needed.
acyclovir, ampicillin diazepam, gancyclovir heparin, imipenem phenytoin	$NaHCO_3$ (8.4%) or NaOH (0.1 mmol/L)	
Acidic Drug Precipitates		
calcium gluconate, ceftriaxone, PN mixtures, vancomycin	HCL (0.1 N) or L-cysteine (50 mg/mL)	
Mineral Precipitates		
PN mixtures rich in Ca^{++} and PO_4^-	NaOH (0.1 N) for first attempt, then L-cysteine (50 mg/mL)	
Lipid or Protein Residues		
PN mixtures, Propofol	ETOH (70%) for lipids NaOH (0.1 N) for lipids and proteins	1. Instill 10 mL over 60 min. 2. Dwell for 2 hrs. 3. Attempt flushing with 10–20 mL 0.9% saline. 4. Repeat if needed.

From References 11 and 14. ETOH = ethanol, HCL = hydrochloric acid, NaHCO3 = sodium bicarbonate, NaOH = sodium hydroxide, PN = parenteral nutrition.

NON-THROMBOTIC OCCLUSION: Failure of the thrombolytic regimen to restore patency should prompt consideration of the non-thrombotic causes of catheter occlusion shown in Table 3.3. Note that alkaline precipitates are cleared with an alkaline solution (sodium bicarbonate or sodium hydroxide) and acidic precipitates are cleared with an acid (hydrochloric acid), while lipid residues can be cleared with 70% ethanol (11,14).

NON-INFECTIOUS COMPLICATIONS

The non-infectious complications of indwelling catheters include catheter-related venous thrombosis and vascular perforation.

Catheter-Related Thrombosis

Indwelling catheters can trigger localized thrombus formation by injuring the vascular endothelium, or by platelet adhesion to the catheter. Catheter-related thrombosis is especially prevalent with peripherally inserted central catheters (PICCs), but the thrombosis is clinically silent in more than 95% of cases (18). There is an increased risk of catheter-related thrombosis in cancer patients, especially those with long-term, tunneled catheters for chemotherapy (19).

Upper Extremity Thrombosis

Deep vein thrombosis (DVT) is not limited to the lower extremities; i.e., about 5% of cases involve the deep veins above the elbow (19,20), and as many as 90% of these are attributed to the presence of a catheter (e.g., a midline catheter or PICC) (20). The other risk factor for this type of DVT is malignancy; i.e., one-quarter to one-third of patients with upper extremity DVT have an active malignancy (19).

Upper extremity DVT can be clinically silent, and the only consequence may be difficulty aspirating blood through the involved catheter. Complete thrombotic obstruction eventually produces swelling of the upper arm, which can be accompanied by discomfort, paresthesias, arm weakness, and engorgement of the superficial veins around the shoulder (Urschel's sign!) (19). The thrombosis can also propagate into the superior vena cava, but the *superior vena cava syndrome* (with facial swelling, headache, etc.) is a rare consequence of catheter-associated DVT (21). Finally, fewer than 10% of upper extremity DVTs are associated with symptomatic pulmonary emboli (22).

DIAGNOSIS: Compression ultrasonography is the diagnostic test of choice for upper extremity DVT (see Figure 1.7 for a normal compression test). A positive test (i.e., clot-filled veins are not compressible) has s sensitivity of 97% and a specificity of 96% for upper extremity DVT (19). D-dimer levels are not reliable as a screening test for DVT because critically ill patients often have elevated D-dimer levels.

MANAGEMENT: Removal of the offending catheter is not mandatory in upper extremity DVT, and is recommended only when arm swelling is severe or painful, or when anticoagulant therapy is contraindicated (19,20). The anticoagulant regimens used for lower extremity DVT have been adopted for the upper extremity. These regimens are described in Chapter 22.

Lower Extremity Thrombosis

Femoral vein catheters are considered high-risk for catheter-related thrombosis, especially when left in place for several days. However, the reported incidence of symptomatic DVT with these catheters is only 1.6% (23), and there is no documented correlation between the duration of catheter use and the appearance of DVT (23). Therefore, the risk of DVT with these catheters is overstated.

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Vascular Perforation

Perforation of the superior vena cava or right atrium is an uncommon but potentially life-threatening complication of central venous cannulation. This complication is also avoidable with vigilance for malpositioned catheters.

Superior Vena Cava Perforation

Left-sided central venous catheters that are advanced into the superior vena cava (SVC) do not always make the acute turn downward toward the right atrium. When this occurs, the tip of the catheter abuts the lateral wall of the SVC, as shown in the upper chest x-ray in Figure 3.1. If the catheter remains in this position, the tip of the catheter can erode through the vessel wall, resulting in leakage of blood and intravenous fluids into the mediastinum and pleural space. The symptoms from SVC perforation i.e. (substernal chest pain, cough, and dyspnea) are nonspecific, and suspicion of perforation is often prompted by the sudden appearance of mediastinal widening and a pleural effusion on the chest x-ray, like the lower x-ray in Figure 3.1 (24).

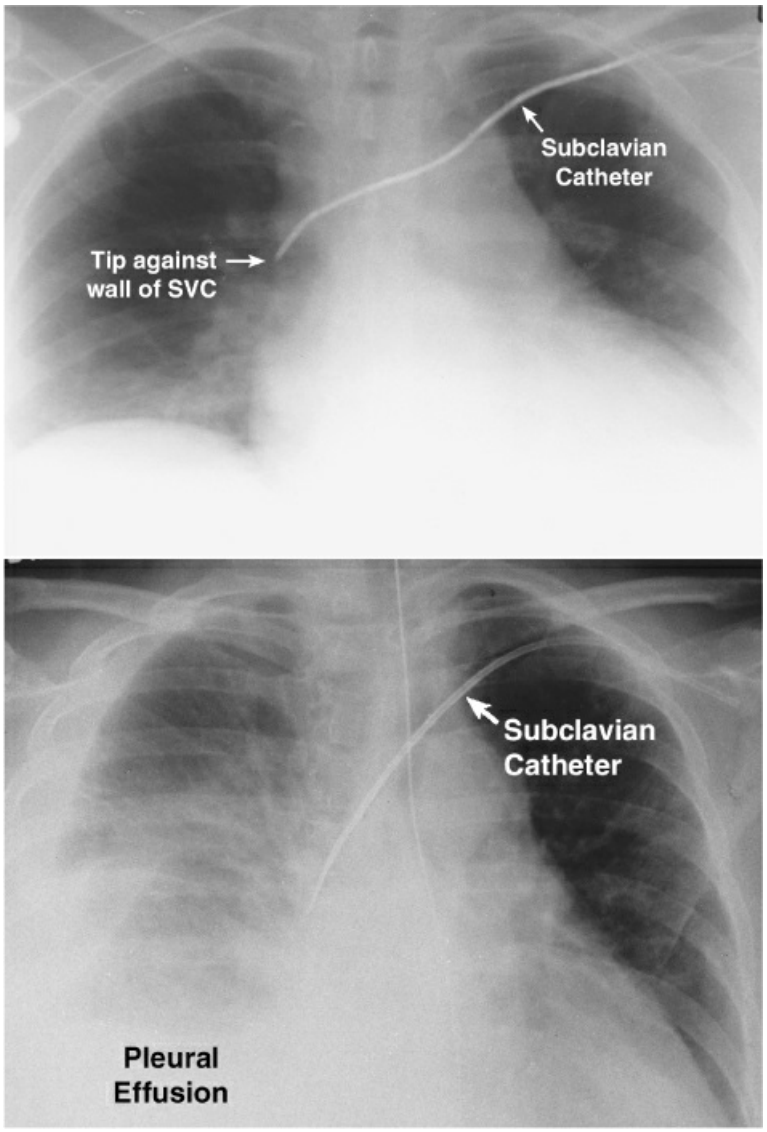


FIGURE 3.1 Chest x-rays showing a subclavian vein catheter that is positioned to perforate the superior vena cava (upper panel), and a similarly positioned catheter that has perforated the vena cava (lower panel). Lower chest x-ray from Reference 24.

DIAGNOSIS: The pleural effusions can be largely composed of intravenous fluids that escape through the perforation; as such, a thoracentesis can support the diagnosis of SVC perforation if the pleural fluid is similar in composition to the intravenous infusion fluid. The perforation is confirmed by injecting radiocontrast dye through the catheter and noting the presence of dye in the mediastinum or pleural space.

MANAGEMENT: When vena cava perforation is first suspected, infusion through the catheter should be stopped immediately. If the diagnosis is confirmed, the catheter should be removed (this does not provoke mediastinal bleeding) (24). Antibiotic therapy is not necessary unless there is evidence of infection in the pleural fluid (24).

Cardiac Tamponade

The most life-threatening complication of central venous catheters is cardiac tamponade from a catheter tip that perforates the right atrium. Fortunately, this complication is rare (see References 44 and 45 in Chapter 1). The first sign of tamponade is usually the abrupt onset of dyspnea, which can progress to cardiovascular collapse within an hour. The diagnosis requires ultrasound evidence of a pericardial effusion with diastolic collapse of the right heart, and immediate pericardiocentesis is necessary to reduce the pericardial pressure and allow the right heart to fill. Emergency thoracotomy may also be necessary if there is a large tear in the wall of the heart. The mortality rate in this condition varies from 40% to 100% in published reports (25).

CATHETER-RELATED BLOODSTREAM INFECTIONS

Indwelling vascular catheters (with the exception of short, peripheral vein catheters) are the leading source of nosocomial bloodstream infections, and septicemia from indwelling catheters has a mortality rate of 12–25% in ICU patients (26,27). The average rate of infections involving central venous access catheters is 4.6 per 1,000 catheter days (28).

Pathogenesis

Colonization of indwelling vascular catheters has several potential sources, as indicated in Figure 3.2. Each of these sources is described below using the corresponding numbers in the illustration.

. Microbes can gain access to the bloodstream via contaminated infusates (e.g., blood products), but this occurs rarely.

. Contamination of the internal lumen of vascular catheters can occur through break points in the infusion system, such as catheter hubs. This may be a prominent route of infection for catheters inserted through subcutaneous tunnels.

. Microbes on the skin can migrate along the subcutaneous tract of an indwelling catheter. This is considered the principal route of infection for percutaneous (non-tunneled) catheters, which includes most of the catheters used in the ICU. (For a novel view concerning this route of

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infection, see A FINAL WORD at the very end of the chapter).

. Microorganisms in circulating blood can attach to the intravascular portion of the catheter. The catheter is not considered the primary source of infection in this situation, but it is possible that the proliferation of the microbes on the catheter reaches a point where the catheter becomes a source of bacteremia.

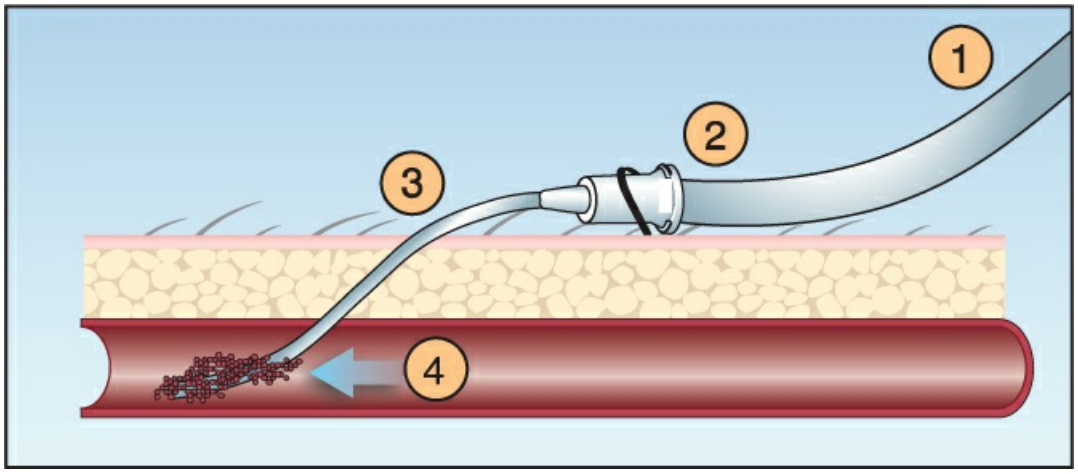


FIGURE 3.2 Sources of microbial colonization of intravascular catheters. See text for explanation.

Biofilms

Microbes have a propensity to congregate on inert surfaces. Once in contact with a surface, they releases adhesive molecules (called *adhesins*, of course) that firmly attach them to the surface. They then begin secreting an extracellular matrix (known as *slime*) that consists of polysaccharides, proteins, fibrin, and extracellular DNA (29). This extracellular matrix creates a protective environment called a *biofilm* that allows microbes to thrive and proliferate without being damaged by the surrounding environment. Phagocytic cells are unable to ingest organisms that are embedded in a biofilm, and *antibiotic concentrations that eradicate free-living bacteria must be 100 to 1,000 times higher to eradicate bacteria in biofilms* (30).

Biofilms are ubiquitous in nature, and predominate on surfaces that are exposed to moisture. (The slippery film that covers rocks in a stream is a familiar example of a biofilm.) They also form on indwelling medical devices such as endotracheal tubes, urinary catheters, and vascular catheters, and can be found on biological surfaces such as the oral mucosa, the epithelial surface in the airways, the mucosal lining of the GI tract, and the surface of native heart valves.

CATHETER-RELATED INFECTIONS: Biofilms play an important role in catheter-related infections, as shown by the following observations:

. Examination of indwelling vascular catheters with scanning electron microscopy has revealed extensive biofilm formation on the surface of the catheters, including those that were in place for only one day (31).

. Coagulase-negative staphylococci, which are the leading cause of catheter-related infections, readily adhere to polymer surfaces and produce slime (32). A biofilm of *Staphylococcus*

epidermidis is shown in Figure 3.3.

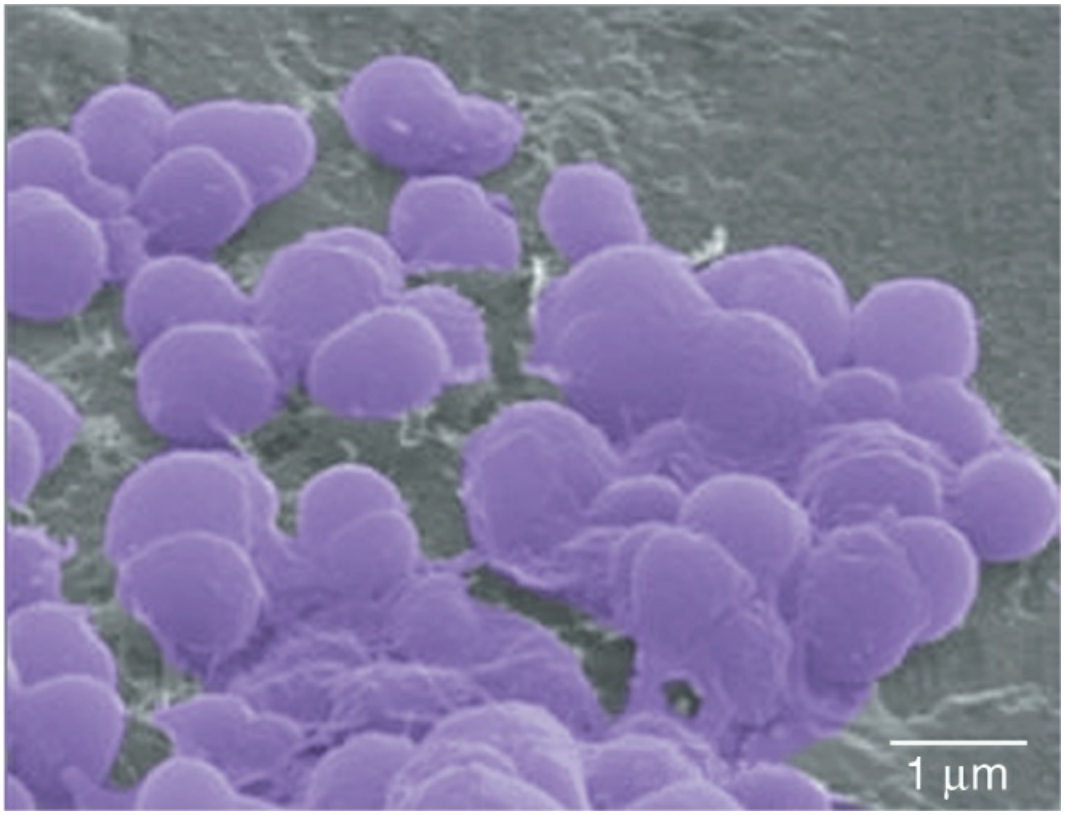


FIGURE 3.3 Electron micrograph of *Staphylococcus epidermidis* encased in a biofilm. Image courtesy of Jeanne VanBriesen, Ph.D., Carnegie Mellon University. Image colorized digitally.

The ubiquitous presence of biofilms on indwelling vascular catheters means that antibiotic therapy is not the optimal treatment for catheter-related bloodstream infections. (The doses of antibiotics needed to achieve bactericidal levels in biofilms would be toxic to the host.) Several anti-biofilm strategies are being evaluated (33), but nothing has reached clinical testing at this time.

Clinical Features

Catheter-related infections do not appear in the first 48 hours after catheter insertion (which presumably is the time required for colonization of the catheter tip). When they do appear, the clinical manifestations are non-specific: e.g.,

. Fever and leukocytosis are common manifestations, but these are signs of inflammation, and are not evidence of infection.

. There may be erythema around the catheter insertion site, but this has no predictive value for identifying catheter-related infections (9).

. Purulent drainage from the catheter insertion site is evidence of infection, but it can be an exit-site infection, without invasion of the bloodstream (34).

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Because of the non-specific nature of these clinical findings, the diagnosis of catheter-related infections is not possible on clinical grounds.

Diagnosis

The diagnosis of catheter-related septicemia requires a positive blood culture from a site other than the catheter, plus some evidence that the catheter is the source of the positive blood culture (and is not secondarily seeded). This evidence was originally provided by a semiquantitative culture of the catheter tip, but this method fell from favor because it required removal of the catheter (which is not desirable, because many cases of suspected catheter-related infection are not confirmed), and the outer surface of the catheter was cultured, thereby missing infections arising from the lumen(s) of the catheter. The culture methods that are currently recommended are summarized in Table 3.4 (35,36).

TABLE 3.4 Recommended Culture Methods for the Diagnosis of Catheter-Related Septicemia

Culture Method	Diagnostic Criteria
Differential Quantitative Blood Cultures	Same organism in peripheral blood & catheter blood, and colony count from catheter blood ≥3 times greater than colony count from peripheral blood. (Accuracy = 94%)[1]
Differential Time to Positive Cultures	Same organism in peripheral blood & catheter blood, and onset of growth in catheter blood at least 2 hours before onset of growth in peripheral blood. (Accuracy = 91%)[1].

From the clinical practice guidelines in Reference 34.

[1] From Reference 37.

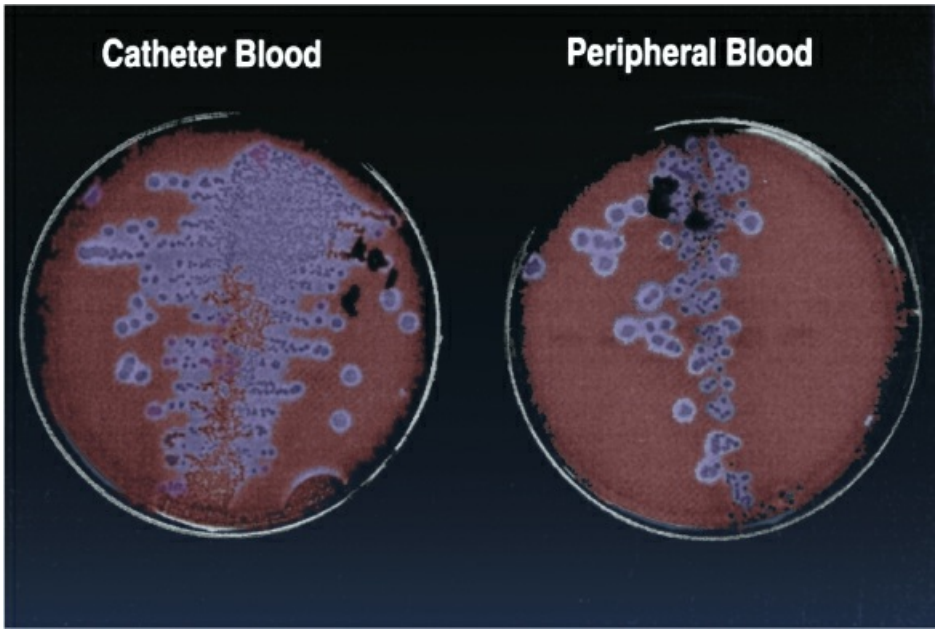


FIGURE 3.4 Culture plates from a case of catheter-related septicemia showing denser growth in blood drawn through a central venous catheter *(Catheter Blood)* compared to blood from a

peripheral venipuncture *(Peripheral Blood)*. From Reference 36. Image colorized digitally.

Quantitative Culture Method

This method uses paired blood cultures (one drawn through the catheter, and one from a peripheral venipuncture site), and is based on the expectation that if the catheter is the source of a bloodstream infection, blood drawn through the catheter will have a higher growth density than the peripheral blood. This requires a "quantitative" blood culture, where growth is expressed as the number of colony forming units per mL (like urine cultures). The diagnosis of catheter-related infection is confirmed if the same organism is isolated from catheter blood and peripheral blood, and the colony count in catheter blood is at least 3 times greater than the colony count in peripheral blood. An example of the comparative growth density in a case of catheter-related septicemia is shown in Figure 3.4.

Timed Culture Method

This method also uses paired blood cultures (one drawn through the catheter, and one from a peripheral venipuncture site), and is based on the expectation that if the catheter is the source of a bloodstream infection, the blood withdrawn through the catheter will show microbial growth at an earlier time than peripheral blood. This method uses routine (qualitative) blood cultures, and the diagnosis of catheter-related infection is confirmed if the same organism is isolated from the catheter blood and peripheral blood, and growth is detected at least 2 hours earlier in the catheter blood.

Which Method is Preferred?

Neither method is clearly superior to the other. The quantitative method has a sightly higher accuracy (37), but the difference is small (94% versus 91%), and may not justify the higher cost of the quantitative method. There is actually little enthusiasm for either culture method, and the diagnosis of catheter-related infection is often based on the same organism being isolated in catheter blood and peripheral blood, with no other apparent source of infection.

HOW MANY LUMENS TO CULTURE? Central venous catheters typically have three lumens (and can have four) and each lumen is a potential source of septicemia (38). However, the recommended practice is to obtain catheter-derived blood cultures from a single lumen (34). The problem with this practice is highlighted by a study showing that *38% of catheter-related infections would be missed if single-lumen cultures were used in the diagnostic evaluation* (39). Using a pooled blood culture from all lumens (to reduce the volume of blood needed to culture multiple lumens) has not proven to be as sensitive as cultures from each lumen (40). At the present time, the available evidence indicates that *the diagnostic approach to catheter-related infection is optimal when blood cultures are drawn through each lumen of the catheter*.

TABLE 3.5 The Top Ten Pathogens in Catheter-Related Septicemia

1–5	6–10
1. Coag-negative staphylococci (16%)†	6. *Candida albicans* (6%)
2. *Staphylococcus aureus* (13%)	7. *Escherichia coli* (5.4%)

3. *Enterococcus faecalis* (8.4%)	8. Other *Candida* spp. (5%)
4. *Klebsiella pneumoniae* (8.4%)	9. *Enterobacter* spp. (4.4%)
5. *Enterococcus faecium* (7%)	10. *Pseudomonas aeruginosa* (4%)

†Numbers in parentheses represent percentage of all isolates.
From Reference 41.

Management

Empiric antibiotics are recommended for all ICU patients with suspected catheter-related infection, and they should be started immediately after cultures are obtained. The pathogens that are most frequently involved in catheter-related infections are shown in Table 3.5. This list is from the National Healthcare Safety Network (41), and it represents the summed contribution of 4,515 hospitals. Note that staphylococci and enterococci together account for 45% of all offending pathogens.

Empiric Antibiotic Therapy

The recommendations for empiric antibiotic coverage are shown in Table 3.6. Van comycin is the backbone of the empiric regimen because it is the most active agent
against staphylococci (including coagulase-negative and methicillin-resistant strains),
and enterococci. Daptomycin can substitute for vancomycin if there is evidence of van comycin-resistance. Empiric coverage for gram-negative enteric organisms is advised for neutropenic or seriously ill patients (35); suitable antibiotics include the carbapenems (e.g., meropenem), the fourth-generation cephalosporins (e.g., cefepime), and the β-lactam/β-lactamase inhibitor combinations (e.g., piperacillin/tazobactam). Additional gram-negative coverage (with an aminoglycoside) is recommended when multidrug-resistant gram-negative organisms are possible offenders

Empiric coverage for candidemia is recommended when the conditions listed in Table 3.6 are present. The echinocandins (caspofungin, micafungin, and anidulafungin) are more active than the azoles (e.g., fluconazole) against all *Candida* species and thus are favored for empiric antifungal coverage (35).

Catheter Management

As mentioned earlier, most cases of suspected catheter-related infections are not con firmed by culture results: e.g., in one study, 1,527 catheters were removed for suspected catheter-related infection, and the diagnosis was confirmed in only 169 (11%) of the cases (28). Therefore, catheters should be left in place pending culture results.

TABLE 3.6 Empiric Antibiotic Coverage for Common Pathogens in Catheter-Related Infections

Organism	Antibiotic	Comment
Staphylococci	Vancomycin	If MRSA isolates with MIC >2 mg/mL are prevalent, use daptomycin..
Enterococci	Vancomycin	If vancomycin resistance is a concern, use daptomycin.

Gram Negative Bacilli	Carbapenem[a] or Cefepime or Piperacillin-Tazobactam	Add aminoglycoside for neutropenia or concern for multidrug-resistant organisms.
Candida species	Echinocandin[b]	***Indications:*** Femoral catheter, TPN, hematologic malignancy, prolonged antibiotic Rx, recent transplant, or *Candida* spp. elsewhere.

From References 34,35.

[a]Carbapenems include imipenem, meropenem, and doripenem.

[b] Echinocandins include caspofungin, micafungin, and anidulafungin.

INDICATIONS FOR CATHETER REMOVAL: If the diagnosis of catheter-related infection is confirmed, catheters should be removed if any of the following is present:

. Hemodynamic instability or progressive multiorgan dysfunction.

. Evidence of endocarditis or septic thrombophlebitis.

. Infections with *Staphylococcus aureus, Candida* species, *Pseudomonas* species, or multidrug-resistant gram-negative bacilli.

. Persistent bacteremia for >96 hrs despite appropriate antimicrobial therapy.

Catheters should not be replaced over a guidewire, but instead should be removed and reinserted at a new venipuncture site. However, guidewire replacement may be the best option if there is very limited venous access.

There is an increasing tendency to leave catheters in place, because inserting a new catheter is time consuming, costly, and can be risky. Even guidewire exchanges can have complications (7). When a catheter is left in place, systemic antibiotics will have limited or no efficacy in decontaminating the catheter (because of the biofilm on the catheter), and recurrent infections are expected (42). In this situation, antibiotic lock therapy should be considered.

ANTIBIOTIC LOCK THERAPY: Instilling concentrated antibiotic solutions into indwelling catheters and allowing an extended dwell time will enhance the ability to disrupt biofilms and eradicate persistent organisms (35). The antibiotic "lock" solution contains the same antibiotic used systemically, in a concentration of 2–5 mg/mL in heparinized saline. This solution is injected into each lumen of the indwelling catheter and allowed to dwell for 24 hours, and is replaced every 24 hours for the duration of the systemic antibiotic therapy. If the catheter is never idle and antibiotic lock therapy is not possible, the systemic antibiotic(s) should be delivered through the suspect lumen.

Duration of Antimicrobial Treatment

The duration of antibiotic therapy is determined by several factors, including the offending pathogen, the status of the catheter (i.e., replaced or retained), and the response to antimicrobial therapy. The following are some general suggestions (34):

. If the infection involves a microbe with limited pathogenicity (e.g., coagulase-negative staphylococci), and there is a favorable response to antimicrobial therapy within 72 hours, then

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no more than 7 days of treatment is necessary (34).

. For cases of *Staph aureus* bacteremia, the accepted practice is to look for evidence of endocarditis with transesophageal ultrasound (the rationale for this is stated later). If there is no evidence of endocarditis, antibiotic therapy can be limited to 14 days if the following conditions are satisfied: the catheter has been removed, the patient is not immunocompromised, and there are no intravascular prosthetic devices in place (34). Evidence of endocarditis mandates 4–6 weeks of antimicrobial therapy, and an infectious disease consult.

. For infections caused by enterococci or gram-negative bacilli, 7–14 days of antibiotic therapy is advised, regardless of whether the catheter is replaced or retained (34).

Persistent Sepsis

Continued signs of sepsis or persistent septicemia after 72 hours of antimicrobial therapy should prompt a search for the following conditions.

Suppurative Thrombophlebitis

The thrombi that form around indwelling vascular catheters can trap microbes from a colonized catheter. These microbes can then proliferate (since blood is a good culture medium) and transform the thrombus into an intravascular abscess. This condition is known as *suppurative thrombophlebitis*, and the most common offending organism is *Staphylococcus aureus* (34). Clinical manifestations are often absent, but can include purulent drainage from the catheter insertion site, limb swelling from thrombotic venous occlusion, and multiple cavitary lesions in the lungs from septic emboli.

The diagnosis of septic thrombophlebitis requires evidence of thrombosis in the cannulated blood vessel (by ultrasound) and persistent septicemia with no other apparent source. Treatment includes catheter removal and systemic antibiotic therapy for 4–6 weeks (2). Surgical excision of the infected thrombus is usually not necessary, and is reserved for cases of refractory septicemia. There is no consensus on the use of heparin anticoagulation in suppurative thrombophlebitis, and current guidelines (34) recommend that heparin therapy should be considered.

Endocarditis

Nosocomial endocarditis is uncommon, with the reported incidence in university teaching hospitals being 2–3 cases annually (43,44). Vascular catheters are implicated in 30 to 50% of cases, and staphylococci (mostly *S. aureus*) are the offending organisms in up to 75% of cases (43,44). Methicillin-resistant strains of *S. aureus* (MRSA) predominate in some reports (45).

Typical manifestations of endocarditis (e.g., new or changing cardiac murmur) can be absent in as many as two-thirds of patients with nosocomial endocarditis involving *Staphylococcus aureus* (45), which is why a search for endocarditis (with transesophageal ultrasound) is advised for all patients with *S. aureus* bacteremia.

A FINAL WORD

A Novel View of Catheter-Related Infections

One of the core beliefs in critical care is the notion that catheter-related bloodstream infections are caused by skin microbes that travel down the tract of the catheter and colonize the distal end of the catheter. This paradigm for catheter-related infections is rarely questioned, yet there are a number of observations that do not fit the model. The following observations suggest that the skin is *not* the source of catheter colonization and infection.

. There is a poor correlation between the organisms found on the skin at the catheter insertion site, and the organisms found on the distal end of central venous catheters (46).

. Decontamination of the catheter insertion site with antibiotic ointments does not reduce the incidence of catheter-related bloodstream infections (3).

. The shortest route for skin microbes to enter the bloodstream is via peripheral vein catheters, which sit in superficial veins, yet these catheters are rarely implicated as a cause of catheter-related septicemia.

The next set of observations suggest that the bowel is the source of catheter-related infections:

. Gram-negative bacilli, enterococci, and *Candida* species account for about half of the microbes that are responsible for catheter-related infections (see Table 3.5), and these organisms are inhabitants of the bowel, not the skin.

. In cases of catheter-related infections, cultures of the upper GI tract show a better correlation with catheter tip cultures than cultures of the skin around the catheter insertion site (47).

. Decontamination of the bowel with nonabsorbable antibiotics has been shown to markedly reduce the incidence of catheter-related infections, and in some studies, these infections were completely abolished by bowel decontamination (48).

The prominence of staphylococci in catheter-related infections does not eliminate the bowel as a source of these infections because staphylococci are prominent inhabitants of the bowel during prolonged antibiotic therapy (49), and in critically ill patients (50). In fact *Staphylococcus epidermidis* (the leading cause of catheter-related infections) *is one of the most prevalent organisms in the upper GI tract of patients with multiorgan failure* (50). The species name *epidermidis* is thus a misnomer, since these organisms do not exclusively inhabit the epidermis.

The catheters that are responsible for catheter-related infections are lengthy (i.e., 6 to 12 inches for central venous catheters, and 15 to 20 inches for peripherally inserted central catheters) and the long intravascular segments of these catheters may be prone to seeding during occult or transient bacteremias. Once seeded, microbial proliferation on the catheter could reach the point where the catheter seeds the bloodstream, creating the appearance of a primary catheter-related bloodstream infection. Seeding of the catheters with bowel organisms is very possible, because the bowel is considered a prominent source of occult sepsis in critically ill patients (50). (See Chapter 4.)

Why does this matter? Because if the skin is not the site of origin for catheter-related infections, then we are spending a lot of time and money decontaminating the wrong surface.

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A Final Word

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A L G r a w a n y

Section II

COMMON PRACTICES

We think so because all other people think so;
or because we think we in fact do think so;
or because we were told to think so and think we must think so;
or because we once thought so and think we must think so
or because of having thought so, we think we must still think so.

Henry Sidgwick
(1838–1900)

Chapter 4

Alimentary Prophylaxis

We are told the most fantastic biological tales. For example, that it is dangerous to have acid in your stomach.

JBS Haldane (1939)

Antiseptic practices are focused almost exclusively on the skin, but there is another, even larger surface that can be breached by microbes: i.e., the mucosal lining of the gastrointestinal (GI) tract. The lumen of the GI tract is outside the body (like the hole in a donut), and the mucosal surface serves as a barrier to microbial invasion, just like the skin. However, the skin is a multilayer affair with a keratinized covering, while the GI mucosa (our "inner skin") is a single layer of columnar epithelial cells that is only 0.1 mm thick, and serves as a barrier for an estimated 100 to 400 *trillion* microbes that inhabit the bowel (1). This scenario of a paper-thin barrier for an enormous army of microbes highlights the threat of microbial invasion from the bowel.

This chapter will introduce you to the importance of the alimentary tract (which extends from the oral cavity to the rectum) as a source of infection in critically ill patients, and describes the antiseptic measures that are available for this site. Also included is a section on stress-related mucosal injury in the stomach, and what can be done to prevent troublesome bleeding from this condition.

MICROBIAL INVASION FROM THE BOWEL

The Intestinal Microbiome

Bacteria are aquatic in nature, and the moisture-rich environment in the alimentary tract is ideal for microbial proliferation. The GI tract is home to about 70,000 distinct bacterial genomes (2), with an estimated population of $100 – 400 \times 10^{12}$ organisms (1), and a total mass of about 2 kg (4.4 pounds) (3). The distribution of this mass is not uniform, as depicted in Figure 4.1 (4). Note that the stomach is the least populated region of the alimentary tract, with fewer than 1,000 organisms per mL of gastric secretions. Overall, about 99% of the microorganisms in the GI tract are located in the large bowel.

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The microbial population in the bowel is present almost from birth (presumably introduced in the mother's milk), and it functions as a collective "ecosystem" that is known as a *microbiome*. The intestinal microbiome is protected from invaders by the actions of gastric acid, which eradicates pathogens ingested in contaminated food products (e.g., *Salmonella* spp), or transmitted via the fecal-oral route (e.g., *Clostridium difficile*). (The antimicrobial actions of gastric acid are described later.)

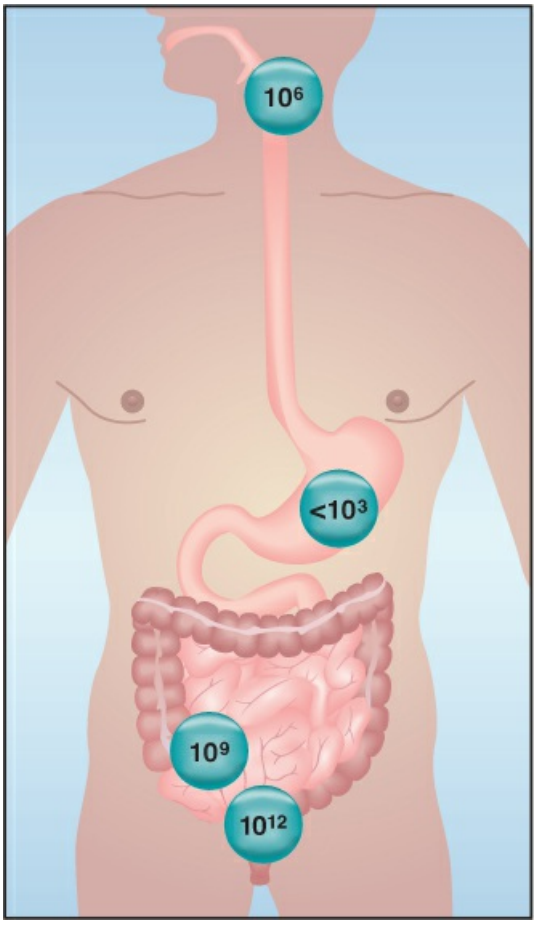


FIGURE 4.1 The population density of microorganisms in different regions of the alimentary tract. Numbers indicate colony forming units per mL of luminal contents. (From Reference 4).

Protective Effects

The intestinal microbiome has a symbiotic relationship with the host organism, and is instrumental in protecting against microbial invasion from the bowel. This is achieved by the following mechanisms (2,5):

. Resident microbes prevent colonization with pathogenic organisms by attaching to the intestinal mucosa and occupying all available spaces.

. Microbial production of short-chain fatty acids provides essential fuels for the mucosal cells in the bowel, which helps to maintain the mucosal barrier.

. Microbial products contribute to the proper functioning of the gut-associated lymphoid tissue

(GALT) which traps and destroys microbes that breach the mucosal barrier (6).

Disruption of the microbiome, a condition known as *dysbiosis*, impairs each of these mechanisms and promotes pathogenic invasion from the bowel. This is illustrated in Figure 4.2. The movement of pathogens across the bowel wall is known as *translocation* (7), and it is a principal source of sepsis and multiorgan failure in critically ill patients.

Multiorgan Failure

The leading cause of death in critically ill patients is multiorgan failure (8), a condition characterized by systemic inflammation (fever, leukocytosis, etc.) that persists or progresses, and is accompanied by progressive dysfunction in two or more major organs (9). The bowel is believed to be the source of this life-threatening condition (10), as described next.

The Motor of Multiorgan Failure

The inciting event in multiorgan failure can be a derangement of the intestinal microbiome (dysbiosis), or a period of splanchnic hypoperfusion. Both conditions lead to disruption of the gut mucosal barrier, which then permits enteric pathogens and/or proinflammatory mediators to gain access to the systemic circulation. This initiates a systemic inflammatory response (e.g., fever, leukocytosis), which is accompanied by hemodynamic changes (i.e., sympathetic nervous system activation with splanchnic vasoconstriction) that promotes further disruption of the gut mucosal barrier, and so on. The result is a self-sustaining process that continually drives systemic inflammation, and eventually produces inflammatory injury in multiple organs. According to this scenario (and to borrow a popular phrase), *the bowel is the 'motor' of multiorgan failure* (10).

Gastric Acid as an Antiseptic Agent

Gastric acid is often misperceived as a digestive aid. Although an acid environment in the stomach does facilitate the absorption of iron and calcium, patients with achlorhydria (the absence of gastric acidity) are not troubled by nutrient malabsorption. As mentioned earlier, *the principal function of gastric acid is not to facilitate digestion, but to serve as an antimicrobial defense against ingested pathogens* (11). The antimicrobial value of acids has been known since the introduction of antiseptic medicine (see next).

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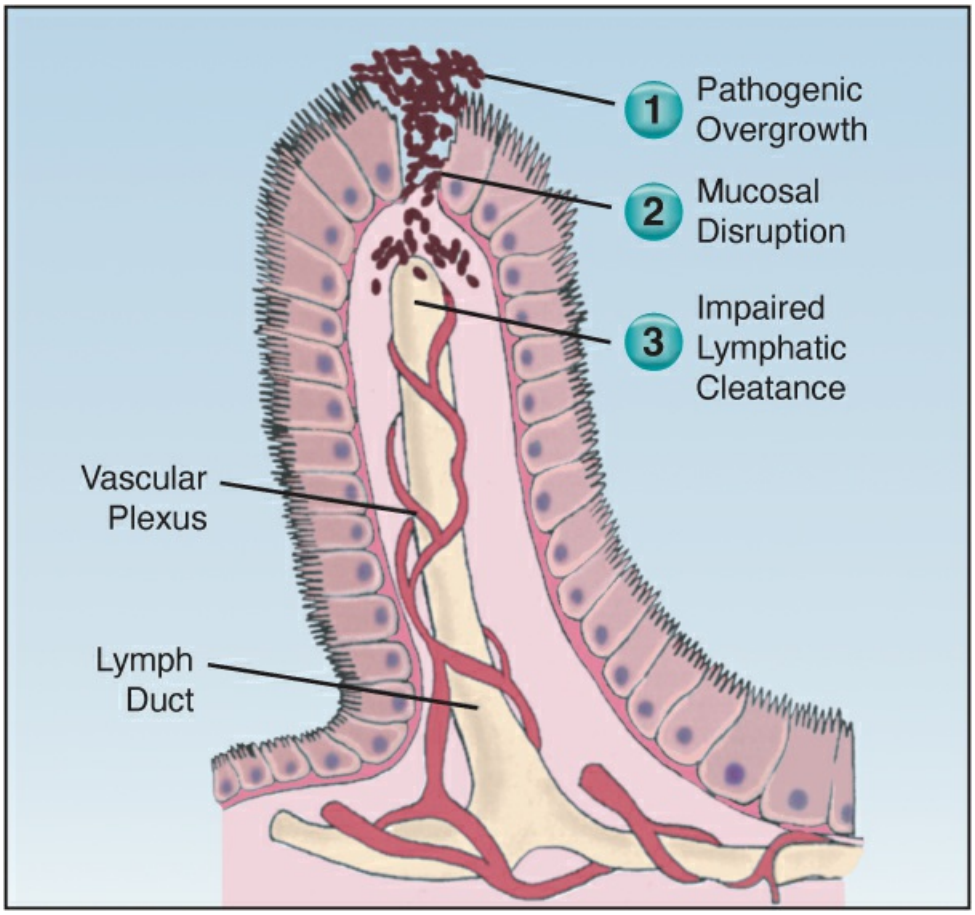


FIGURE 4.2 Consequences of disrupting the intestinal microbiome, which promotes the invasion of pathogens from the bowel.

Historical Note

The benefits of antisepsis were first recognized in the mid-nineteenth century by a British surgeon named Joseph Lister, who discovered that a chemical agent used to treat sewage could also reduce the number of suppurative wound infections. Lister's observations were published in 1867 in a treatise entitled *On the Antiseptic Principle in the Practice of Surgery* (12). In the following excerpt from this treatise, Lister identifies the chemical agent that he used.

> "The material which I have employed is carbolic acid, a volatile compound which appears to exercise a peculiarly destructive influence upon low forms of life, and hence is the most powerful antiseptic with which we are at present acquainted."

As indicated, the first antiseptic agent was an acid (carbolic acid, also known as phenol), so Joseph Lister not only discovered antisepsis, he also discovered the antimicrobial value of acidity. Unfortunately, Joseph Lister's contributions are largely forgotten, and the name Lister is recognized mainly in relation to a popular mouthwash, Listerine®!

Antimicrobial Effects

The influence of gastric pH on the growth of a pathogenic organism is shown in Figure 4.3 (13). The pathogen in this case is *Salmonella typhimurium*, a common cause of infectious enteritis in humans. The graph in Figure 4.3 shows the growth pattern of *S. typhimurium* in gastric

secretions at three different pH levels. Note that the organism thrives at a pH of 4, but there is a steady decline in survival at a pH of 3, and the organism is completely eradicated after one hour at a pH of 2. The pH of gastric secretions is 1–2, so gastric acidity should be very effective in eradicating ingested pathogens.

The role of gastric acidity as an antimicrobial defense mechanism is demonstrated by the consequences of gastric acid suppression (by histamine H_2 receptor antagonists or proton pump inhibitors), which includes an increased risk of infectious gastroenteritis from *Salmonella* and *Campylobacter* species (11,14–16), and an increased incidence of *Clostridium difficile* infections (17). (This topic will resurface later in the chapter.)

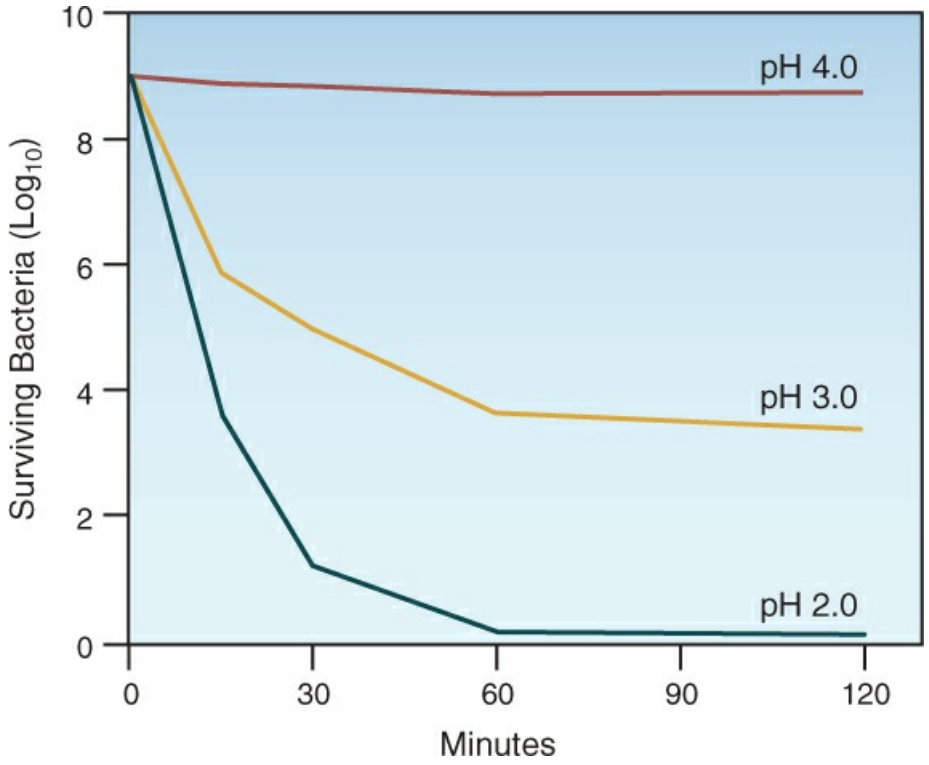


FIGURE 4.3 The influence of gastric pH on the growth of *Salmonella typhimurium*, a common cause of infectious enteritis. (From Reference 13).

ALIMENTARY DECONTAMINATION

Patients who are seriously ill are prone to colonization of the oropharynx and GI tract with pathogenic organisms, and the following decontamination methods are used to limit this process.

Oral Decontamination

The aspiration of mouth secretions into the upper airways is believed to be the inciting event in most cases of hospital-acquired pneumonia (18). The volume of saliva produced daily averages about 750 mL in adults, and each mL of saliva contains about one million microorganisms (see Figure 4.1). Therefore, aspiration of 0.13% (1/750) of the daily saliva volume will introduce a million microbes into the airways. This not a concern in healthy subjects, because the

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oropharynx is colonized with organisms that show little proclivity for harm. Not so in critically ill patients, as described next.

Colonization of the Oropharynx

The oropharynx of seriously ill patients becomes colonized with pathogenic organisms, most notably aerobic gram negative bacilli like *Pseudomonas aeruginosa* (18,19). This colonization is not environmentally driven, but is directly related to the presence and severity of illness. This is demonstrated in Figure 4.4 (19), which shows that healthy subjects are not colonized with aerobic gram-negative bacilli, regardless of the environment.

Pathogenic colonization of the oropharynx is caused by a change in bacterial adherence to the surface epithelium. Epithelial cells have specialized adhesion proteins that bind specific groups or species of bacteria, and in the presence of a serious illness, there is a conformational change in these proteins that results in the ability to bind pathogens. This same process occurs in the bowel mucosa and the bladder epithelium, and it demonstrates that *the determining factor in the colonization of epithelial surfaces is not the presence of an organism, but the ability of the organism to bind to the epithelium*. This topic deserves much more attention, as the prevention of pathogen binding to epithelial surfaces would be an effective method of infection control.

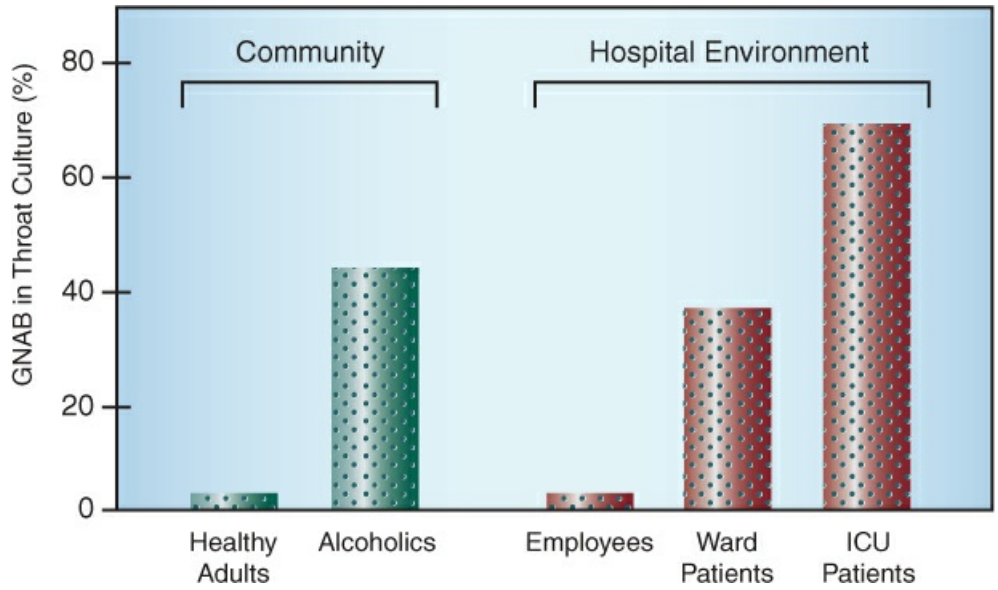


FIGURE 4.4 The prevalence of Gram negative aerobic bacilli (*GNAB*) in cultures of the oral cavity in selected groups of subjects. (From Reference 19).

TABLE 4.1 Oral Decontamination Methods

Method	Regimen
Chlorhexidine	• Apply a 0.12% chlorhexidine gluconate solution to the oral mucosa every 6 hours. • Avoid the 2% chlorhexidine solution, which can cause an ulcerating mucositis.
Selective Oral Decontamination†	• Have pharmacy prepare a gel mixture of 2% gentamicin, 2% colistin, and 2% vancomycin. • Apply this gel to the oral mucosa with a gloved finger every 6 hours.

†Regimen from Reference 27.

Colonization of the oropharynx with gram-negative bacilli is a prelude to pneumonia, since the same organisms (gram-negative bacilli) are the most frequent cause of nosocomial pneumonias (see Chapter 29). This is the rationale for oral decontamination, which is currently reserved for patients who are intubated and receiving mechanical ventilation. There are two methods of oral decontamination: one uses an antiseptic oral rinse, and the other involves the local application of nonabsorbable antibiotics. Both methods are summarized in Table 4.1.

Chlorhexidine

The popular skin antiseptic chlorhexidine has been adopted for use in decontaminating the oropharynx. A 0.12% solution of chlorhexidine gluconate is used as an oral rinse every 6 hours, and this regimen continues for as long as the patient is ventilator-dependent. There is a more concentrated (2%) chlorhexidine gluconate solution, but this is less desirable because it can cause a mucositis (20).

The benefit of oral decontamination with chlorhexidine has been difficult to establish. Some studies have shown a 30% relative reduction in the incidence of ventilator-associated pneumonia, without an associated decrease in mortality (21). However, other studies have shown marginal or no benefit (22), and the pooled results of 11 clinical trials has shown an increase in mortality associated with chlorhexidine (23). Because of concerns about the potential for harm, the international guidelines on ventilator-associated pneumonia withdrew a recommendation for oral decontamination with chlorhexidine (24), and experts in the United States have followed suit (25). Unfortunately, there is no other antiseptic agent that has been adequately studied for oral decontamination.

An additional problem with chlorhexidine that is rarely mentioned is its limited spectrum of activity; i.e. *chlorhexidine is effective primarily against gram-positive organisms (26), while gram-negative organisms are the predominant microbes in the oropharynx of critically ill patients*. This would explain the disappointing results of the clinical trials just mentioned.

Selective Oral Decontamination

The principle behind selective oral decontamination (SOD) is to eradicate colonization with pathogens while preserving the normal microbial population of the oropharynx. This is achieved with a combination of nonabsorbable antibiotics that are locally applied to the oral mucosa. One SOD regimen that has had success in reducing the risk of ventilator-associated pneumonia is shown in Table 4.1 (27). In this case, a mixture of 2% gentamicin, 2% colistin, and 2% vancomycin is applied as a paste to the oral mucosa four times daily. This regimen is designed to eradicate staphylococci, gram-negative aerobic bacilli, and *Candida* species, but has little activity against anaerobic organisms (normal mouth flora).

EFFICACY: Several clinical studies have shown that SOD reduces the incidence of ventilator-associated pneumonia (23,28,29), and also reduces the frequency of bacteremias involving gram-negative bacilli (29,30). This latter observation is demonstrated in Figure 4.5 (29). Both graphs in this figure show a 33% (relative) decrease in the incidence of gram-negative bacteremias associated with SOD.

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Despite convincing evidence that SOD is effective in reducing the frequency of ICU-acquired infections, SOD is largely ignored. On the other hand, despite evidence of limited efficacy and possible harm, chlorhexidine has been the standard method of oral decontamination. This scenario is the antithesis of evidence-based medicine! The rationale for ignoring SOD is the fear of antibiotic resistance; an unfounded fear that is addressed at the end of the next section.

Selective Digestive Decontamination

Selective digestive decontamination (SDD) is based on the same principle as SOD (i.e., prevent colonization with pathogens while preserving the normal microflora), but is applied to the entire alimentary tract. An SDD regimen that has proven successful is shown below (30).

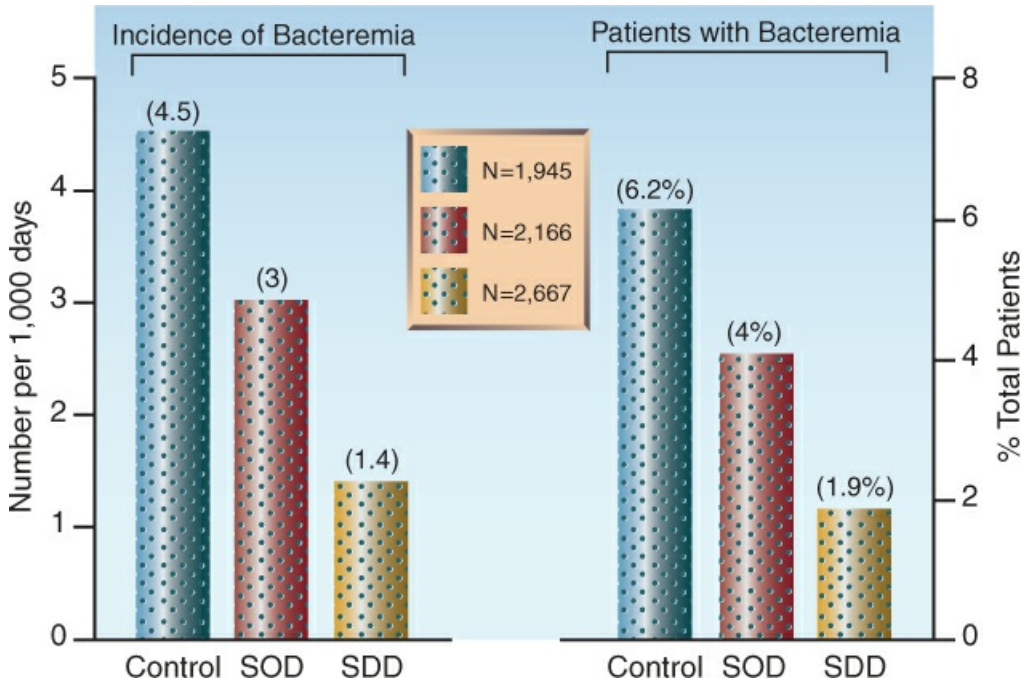


FIGURE 4.5 The prevalence of gram-negative bacteremia in patients randomized to receive selective oral decontamination (SOD), selective digestive decontamination (SDD), or standard care (control). The graph on the left shows the number of bacteremias per 1000 days, and the graph on the right shows the percentage of patients with bacteremia. *N* is the number of patients in each study group. (From Reference 29).

Mouth:	A gel containing 2% polymyxin, 2% tobramycin, and 2% amphotericin is applied to the oral mucosa every 6 hours.
GI tract:	A 10 mL aliquot of a solution containing 100 mg polymyxin E, 80 mg tobramycin, and 500 mg amphotericin is delivered through a nasogastric tube every 6 hours.
Systemic:	Intravenous cefuroxime, 1.5 grams every 8 hours, is used for the first four days.

This nonabsorbable antibiotic combination is designed to eradicate staphylococci, gram-negative aerobic bacilli, and *Candida* species, while sparing the normal microflora of the

oropharynx and GI tract (mainly anaerobes) The intravenous antibiotic is used for systemic protection until decontamination of the bowel is completed, which takes about 5–7 days. The SDD regimen is intended for all patients who will stay in the ICU longer than 72 hours, and is continued until discharge from the ICU.

Efficacy

Numerous clinical studies have shown that SDD reduces the incidence of ICU-acquired infections (including pneumonias, urinary tract infections, and bacteremias), and also significantly reduces the mortality rate (29–34). The impact of SDD on gram-negative bacteremia is demonstrated in Figure 4.5 (29). Note that SDD was associated with a marked (67%) reduction in the frequency of these infections. This effect is lost for bacteremias involving multidrug-resistance gram-negative bacilli (35), indicating that SDD is not appropriate in ICUs with problematic antibiotic resistance.

SOMETHING OF INTEREST: One observation that deserves mention comes from a clinical study where SDD *eliminated* vascular catheter-related bacteremias (32). This suggests that the bowel is the source of catheter-related bacteremias (not the skin); a concept that is presented at the very end of Chapter 3 (see A FINAL WORD).

Why is SDD Ignored?

Despite numerous studies over the past 30 years showing that SDD is an effective method of reducing ICU-acquired infections, SDD is ignored (like SOD). Much of this is the result of the age-old fear that SDD will promote the emergence of antibiotic resistant organisms. However, none of the SDD studies has shown an increase in infections from antibiotic-resistant organisms, including a prospective study evaluating SDD over a 5-year period (36), and a 16-year experience with SDD (37). Thus, the disregard for SDD is not validated by clinical studies.

Probiotics

Probiotics are microbial preparations of normal but harmless inhabitants of the intestinal tract that are capable of preventing colonization with pathogenic organisms. (*Note:* Probiotics should be distinguished from *prebiotics*, which are food substances—mostly dietary fibers—that promote the growth of normal bowel microflora. Probiotics and prebiotics are often combined to create what is known as *synbiotic therapy*.) The most common probiotics are strains of lactic acid bacteria, such as *Lactobacillus* and *Bifidobacterium* species, that are resistant to the bactericidal actions of gastric acid and bile salts, and can successfully colonize the bowel when ingested (38). Another popular probiotic is the *Saccharomyces* fungus, which is the principal ingredient in brewer's yeast and baker's yeast.

Clinical Use

The principal use of probiotics in the ICU is the prevention of *Clostridium difficile* infections during antibiotic therapy (39). This effect is time-limited; the combined results of 19 clinical studies showed that probiotics reduced the risk of *C. difficile* infections only when started within 2 days of antibiotic therapy (40). Probiotic therapy can also reduce the risk of hepatic encephalopathy (41) and ventilator-associated pneumonia (42).

STRESS-RELATED MUCOSAL INJURY

Stress-related mucosal injury is a term used for superficial erosions on the luminal surface of the stomach that are found in critically ill patients. These erosions can be confined to the mucosa, or they can bore deeper into the submucosa. The deeper erosions are called *stress ulcers*, and are more likely to cause troublesome bleeding. Hereafter, both types of surface erosions will be referred to as stress ulcers.

Pathogenesis

The surface epithelial cells in the stomach are covered with a layer of bicarbonate-rich mucus (pH = 7) that protects the cells from the acidity of gastric secretions, and from proteolytic enzymes (e.g., pepsin) in the secretions. Gastric mucosal blood flow plays an important role in preserving the mucous layer by providing nutrients to support the functional integrity of the surface epithelial cells. The importance of mucosal blood flow is demonstrated by the fact that 70–90% of the blood supply to the stomach is delivered to the gastric mucosa (43). The endothelial cells in surface vessels secrete prostacyclin and nitric oxide, which are both instrumental in maintaining nutrient flow to the gastric mucosa (44).

Local ischemia is the inciting event in stress-related mucosal injury, while the presence of luminal acid is believed to aggravate the condition (44). In animal models of hemorrhagic shock, reperfusion injury also plays a role (45). Of interest, occult gastric hypoperfusion can be a frequent occurrence in critically ill patients who are not hypotensive, especially in those receiving mechanical ventilation (44).

Stress Ulcer Bleeding

Erosions are visible on the surface of the stomach in 75% to 100% of patients within 24 hours of admission to the ICU (46). These lesions often ooze blood from eroding into superficial capillaries, but “clinically significant” bleeding is infrequent. The accepted definition of clinically significant bleeding is overt bleeding that is accompanied by at least one of the following: (a) a decrease in blood pressure (systolic, mean, or diastolic pressure) of ≥20 mm Hg, (b) initiation of vasopressor therapy, or a 20% increase in the vasopressor dose, (c) a decrease in hemoglobin of at least 2 g/dL, or (d) transfusion of 2 or more units of packed red blood cells (47,48). The likelihood of significant stress ulcer bleeding is determined by the presence or absence of specific risk factors, as described next.

Risk of Bleeding

The risk of significant bleeding from stress ulcers can be stratified as shown in Table 4.2, which is taken from the most recent clinical practice guidelines on the prevention of stress ulcer bleeding (47). The following points in this table deserve emphasis:

. The risk of bleeding during mechanical ventilation is decreased considerably by enteral tube feedings.

. Corticosteroid therapy is not a significant risk factor for stress ulcer bleeding, and does not require prophylactic therapy. Ditto for therapeutic anticoagulation.

THE STEROID MYTHOS: The popular perception that corticosteroid therapy creates a significant risk of stress ulcer bleeding (and warrants prophylactic therapy) has a surprising lack of experimental support. The most recent evaluation of this issue is a report of the pooled results from 25 studies (49), which showed that corticosteroid therapy had no influence on GI bleeding of any severity. Results like this indicate that corticosteroid therapy should not be considered a risk factor for stress ulcer bleeding.

TABLE 4.2 Identifying the Risk of Significant Stress Ulcer Bleeding

Risk Level[1]	Prophylaxis?	Conditions
Highest Risk (8–10%)	Yes	1. Mechanical ventilation >24 hrs without enteral feedings 2. Chronic liver disease[2]
High Risk (4–8%) (cutoff for prophylaxis)	Yes	1. Worrisome coagulopathy[3] 2. At least 2 conditions from the moderate risk level
Moderate Risk (2–4%)	No	1. Mechanical ventilation >24 hrs with enteral feedings 2. Acute kidney injury 3. Sepsis 4. Shock
Low Risk (1–2%)	No	1. Acute liver failure 2. Anticoagulant therapy 3. Steroid therapy 4. Immunosuppressive therapy 5. Cancer 6. No risk factors

From the clinical practice guideline in Reference 47. See text for definition of significant bleeding.

[1]Numbers in parentheses indicate % patients with significant bleeding.

[2]Cirrhosis with portal hypertension.

[3]Platelets <50 x 10^9/L, INR >1.5, or prothrombin time >20 seconds.

Prophylaxis for Stress Ulcer Bleeding

Surveys indicate that a majority (>80%) of ICU patients receive prophylactic therapy for stress ulcer bleeding, and about 70% of these patients do not need prophylaxis (50). This excessive use of stress ulcer prophylaxis is not only costly, it can also be harmful (see later).

The principal method of prophylaxis for stress ulcer bleeding is the suppression of gastric acid production with proton pump inhibitors (PPIs) or histamine type-2 receptor antagonists (H_2 blockers). (The goal is to maintain a pH ≥4 in gastric aspirates, although this is rarely monitored.) The other, less popular, method employs a cytoprotective agent (sucralfate) that forms a protective covering over damaged areas of the gastric mucosa, and does not alter gastric acidity. The individual drugs and recommended doses are shown in Table 4.3.

TABLE 4.3 Drugs Used for Prophylaxis of Stress Ulcer Bleeding

A L G r a w a n y

Drug	Type	Routes	Dosage
Famotidine	H_2 Blocker	PO, NG, IV	20 mg every 12 hrs†
Pantoprazole	PPI	PO, NG, IV	40 mg once daily
Omeprazole	PPI	PO, NG	40 mg once daily
Lansoprazole	PPI	PO, NG	30 mg once daily
Sucralfate	Protectant	PO, NG	1 g every 6 hrs

†A dose reduction of 50% is recommended for a creatinine clearance <50 mL/min.
PPI = proton pump inhibitor.

Efficacy

All three types of drugs (PPIs, H_2 blockers, and sucralfate) are capable of reducing the risk of stress ulcer bleeding (51). The relative efficacy is: PPIs >H_2 blockers > sucralfate (51–53). However, none of these agents provides a survival benefit (51–53).

H_2 Blockers

Histamine H_2 receptor antagonists, or H_2 blockers, (cimetidine, famotidine) inhibit gastric acid secretion by binding to histamine H_2 receptors on gastric parietal cells. Famotidine is currently the favored H_2 blocker for stress ulcer prophylaxis, and is well suited for ICU patients because it can be given intravenously. The H_2 blockers are cleared by the kidneys, and reduced drug clearance from renal insufficiency can have neurotoxic effects (e.g., altered mentation). For this reason, a 50% reduction in the daily dose is advised for patients with a creatinine clearance <50 mL/min (54).

Proton Pump Inhibitors (PPIs)

The PPIs (omeprazole, esomeprazole, pantoprazole, lansoprazole), are potent acid-suppressing drugs that block the secretion of hydrogen ions by gastric parietal cells. These drugs are actually prodrugs, and are converted to the active form within gastric parietal cells. Once activated, the drugs bind irreversibly to the membrane pump responsible for hydrogen ion secretion, and this results in complete inhibition of gastric acid secretion.

All PPIs are considered equivalent as prophylactic agents (although this has not been studied), but pantoprazole is the most popular PPI in ICU patients because it can be given intravenously (48). (*Note:* Esomeprazole can also be given intravenously, but this drug is rarely used for stress ulcer prophylaxis.) Unlike H_2 blockers, PPIs require no dose adjustment for renal insufficiency.

The graph on the left in Figure 4.6 shows the effect of pantoprazole (40 mg IV once daily) on (significant) stress ulcer bleeding in more than 3,000 high-risk ICU patients (48). Note the low incidence of GI bleeding in the control patients (4.2%), and the small (1.7%) treatment effect. Small benefits like this must be measured against the risks associated with gastric acid suppression (see later).

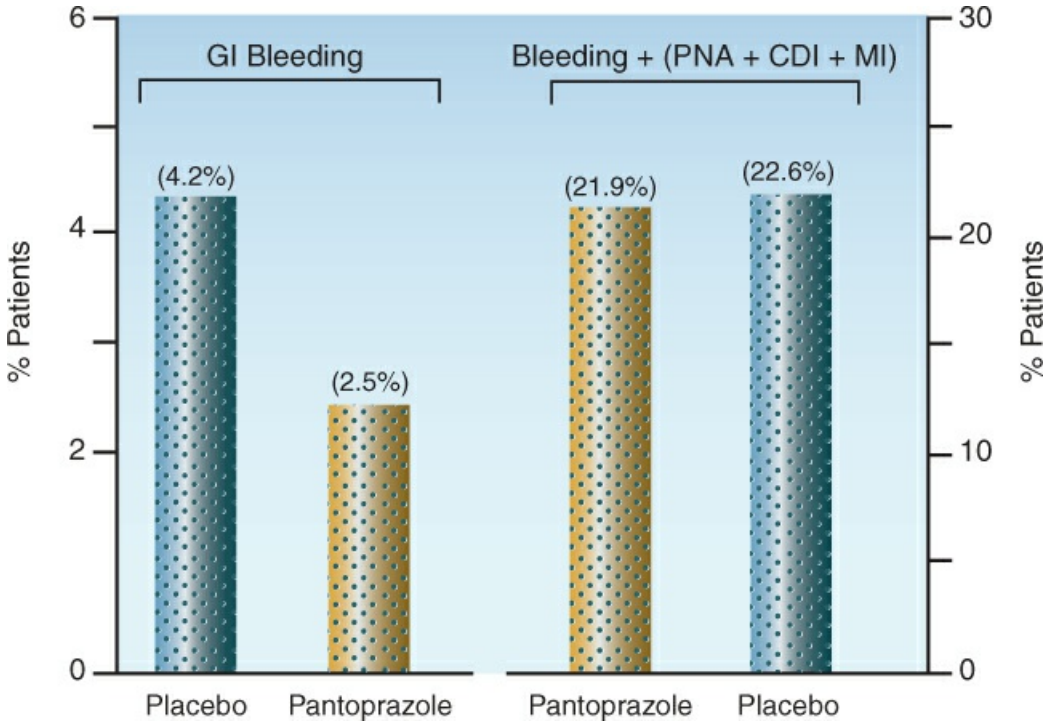


FIGURE 4.6 Effects of stress ulcer prophylaxis with pantoprazole (40 mg IV daily) on the incidence of significant GI bleeding (graph on the left) and the summed incidence of GI bleeding and the risks associated with pantoprazole: pneumonia (PNA), *C. difficile* infection (CDI), and myocardial ischemia (MI)) which is shown in the graph on the right. Data from a study of 3,298 high-risk ICU patients from 6 different countries (48). See text for further explanation.

PPIs AND CLOPIDOGREL: PPIs interact with clopidogrel, an antiplatelet agent that has been used for secondary prevention of acute coronary syndromes and stroke. Clopidogrel is a prodrug that is converted to its active form by the same (cytochrome P-450) pathway in the liver that metabolizes PPIs. As a result, PPIs can impede clopidogrel activation in the liver (by competitive inhibition) and reduce its antiplatelet effect (55). This interaction has been implicated as a risk factor for the recurrence of acute coronary syndromes following percutaneous coronary intervention (56), but evidence of a causal link has been inconsistent. At the present time, there is no recommendation to avoid PPIs in patients receiving clopidogrel.

Risks of Gastric Acid Suppression

The risk of gastric acid suppression is related to loss of the antimicrobial activity of gastric acidity. The major risks are summarized below.

***Clostridium Difficile* INFECTION:** The association between gastric acid suppression and *Clostridium difficile* infections has been reported in outpatients (57), inpatients (58), and ICU patients (59,60), and the association is stronger with PPIs than with H_2 blockers (57,61). Patients who receive PPIs for stress ulcer prophylaxis are three times more likely to develop *C. difficile* enterocolitis (60). In recognition of this risk, The U.S. Food and Drug Administration issued a Drug Safety Alert in 2013 that highlighted the risk of *C. difficile* infections associated with PPIs (62).

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PNEUMONIA: There is also an association between gastric acid suppression and an increased risk of pneumonia, including community-acquired pneumonia (63) and ventilator-associated pneumonia (61). Once again, the risk is greater with PPIs than with H_2 blockers (61). The presumed mechanism is colonization of gastric secretions with pathogenic organisms, and subsequent aspiration of infectious gastric secretions into the upper airways.

Benefits vs. Risks

There is evidence that the benefit of PPIs in reducing stress ulcer bleeding is negated or counterbalanced by the adverse consequences of PPIs. This is demonstrated in Figure 4.6 (48). As mentioned earlier, the graph on the left shows a decrease in the incidence of stress ulcer bleeding associated with a PPI (pantoprazole). The graph on the right shows the composite incidence of stress ulcer bleeding plus the risks associated with PPIs (i.e., *C. difficile* infection, pneumonia, and myocardial ischemia related to the clopidogrel interaction). Note that the total incidence of adverse events is no different in the placebo group and the pantoprazole group. This means that the benefit of pantoprazole prophylaxis is erased if you also consider the adverse consequences of PPI therapy; i.e., there is no net benefit from stress ulcer prophylaxis with PPIs.

Sucralfate

Sucralfate is an aluminum salt of sucrose sulfate that adheres to damaged areas of the gastric mucosa, and forms a viscous covering that shields the denuded surface from luminal acids and pepsin proteolysis. Gastric acidity is maintained, which is the most desirable feature of prophylaxis with sucralfate. At a dose of one gram every 6 hours (usually given as an oral suspension), sucralfate has been shown to reduce the incidence of stress ulcer bleeding (51). Of particular importance, pneumonia is less frequent when sucralfate is used instead of gastric acid-suppressing drugs (64).

Sucralfate has one undesirable feature that precludes its use in a majority of ICU patients: i.e., it can not be used in conjunction with enteral tube feedings. Sucralfate must be given when the stomach is empty (to allow binding to the gastric mucosa), which is not possible during enteral feedings without lengthy interruptions to allow gastric emptying. Even with interruptions, any mixing of sucralfate with tube feedings is problematic because it can promote bezoar formation (65).

Another undesirable feature of sucralfate is the ability to bind certain drugs in the stomach (by virtue of its aluminum moieties), including ciprofloxacin, digoxin, ketoconazole, norfloxacin, phenytoin, thyroxine, tetracycline, theophylline, and warfarin (66). The aluminum in sucralfate can also bind phosphate in the bowel, but this rarely results in hypophosphatemia (67).

The Case Against Stress Ulcer Prophylaxis

The relevant observations about stress ulcer prophylaxis are listed in Table 4.4. These observations indicate that stress ulcer prophylaxis with gastric acid-suppressing drugs provides little or no benefit, and can introduce harm.

The emerging realization that stress ulcer prophylaxis is “more trouble than it’s worth” has prompted efforts to “deprescribe” proton pump inhibitors in high-risk ICU patients. At this time, six clinical trials have shown that withholding pantoprazole in high-risk ICU patients does not increase the occurrence of significant stress ulcer bleeding (68). This differs from earlier studies

showing a decrease in stress ulcer bleeding with PPI prophylaxis, and this difference can be attributed to improvements in enteral nutrition (see next).

Enteral Feeding

Enteral tube feedings can protect against stress ulcer bleeding in two ways. First, enteral feeding has a trophic effect on the gastric mucosa that limits or prevents mucosal breakdown (see Chapter 49). Second, the presence of feeding solutions in the stomach will raise the pH of gastric contents. The prophylactic value of enteral feedings is revealed in the combined results of three clinical studies, which showed that the ability of gastric acid-suppressing drugs to reduce stress ulcer bleeding was lost when patients received a full regimen of enteral tube feedings (69). These results indicate that *enteral feeding is all that is needed for stress ulcer prophylaxis*.

TABLE 4.4 The Case Against Stress Ulcer Prophylaxis

A composite of the following observations indicates that the routine use of stress ulcer prophylaxis (SUP) is not warranted.
• Significant bleeding from stress ulcers is infrequent. • SUP has only a small treatment effect. • There is no survival benefit with SUP. • The benefit of SUP is lost when the risks of gastric acid suppression are considered.

A FINAL WORD

Gastric Acidophobia

One of the recurring themes in this book is the importance of the bowel as a source of ICU-acquired infections, and the risk of these infections is enhanced by the popular (and overly excessive) use of gastric acid suppression for stress ulcer prophylaxis. This practice is rooted in the time-honored fear that acids are corrosive. While acids can corrode certain metals like iron, this does not apply to organic (carbon-based) matter. Consider the following beverages with pH values that are close to gastric acid (70):

Source	pH
Gastric Acid	2.0
Lemon Juice	2.0
Pepsi	2.13
Vinegar	2.2
Coca-Cola	2.24
Fanta	2.51

Despite levels of acidity that are comparable to gastric acid, these beverages are not harmful.

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(Imagine what would happen if Pepsi or Coca-Cola caused mouth burns!) In fact, vinegar is used to *preserve* organic matter (foods); a process known as *pickling*. When an industrial acid causes skin burns, the culprit is the chemical composition of the acid, not the acidity.

We are designed to have acid in our stomach, to protect the intestinal microbiome from foreign invaders, and endowing this acid with the ability to cause harm is contrary to the survival of our species. The cause of stress-related mucosal injury is hypoperfusion of the gastric mucosa, not the acidity of gastric secretions.

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